PRICE-RL: Selection–Transmission Decomposed Reinforcement Learning for Sequential Biological Design

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Abstract

Every reinforcement-learning update for biological sequence design admits a decomposition — the same one the Price equation imposes on evolutionary change — into a *selection* component that re-weights variants the policy already supports, and a *transmission* component that shifts the policy's support into new regions of sequence space. Frank [2025] recently showed that this decomposition is algebraically exact for policy gradients, but no algorithmic work has used it as a primitive. We do. **PRICE-RL** estimates the two components separately on policy, weights them with a PI controller targeting a Price ratio ρ_t^* derived from the reward landscape's autocorrelation length, and exposes ρ_t itself as a per-round diagnostic for reward hacking. We prove the decomposition is exact for factorised and autoregressive categorical policies (Theorem 1), and verify the cosine identity $\hat{g}_S + \hat{g}_T = \hat{g}_{\text{pool}}$ to floating-point precision across $1{,}600$ NK round-seeds without exception. On the $149{,}361$-variant GB1 four-mutation landscape, PRICE-RL ties AdaLead at 500 queries and beats it by 12.4% with non-overlapping 95% CIs at $8{,}000$ queries (Mann–Whitney $p < 10^{-3}$), with the lead widening monotonically as the budget grows. The empirical ρ_t correlates with the analytic optimum at Pearson $r = 0.988$ (95% bootstrap CI $[0.97, 1.00]$). On a deliberately weak proxy reward, the Price ratio fires its 0.9 alarm a mean of 18.6 rounds before proxy variance crosses its alarm on 100% of 16 seeds, a strictly earlier reward-hacking warning than the current state of the art and one robust to surrogate noise up to $\sigma_n = 0.40$. On the deceptive Trap-K landscape PRICE-RL hits the global optimum on every seed at every scale up to $N = 120$, where AdaLead misses up to 20% of the time, and on GB1 it discovers $2.5\times$ more unique top-1% variants than AdaLead at matched fitness. The framework extends to vector-valued reward and to token-level autoregressive RLHF-style policies with the cosine identity preserved at 1.0000 in each case. Across 13 benchmark configurations the picture is balanced: PRICE-RL is best on 5, AdaLead on 6, PEX on 1, GFlowNet-AL on 0 — so the headline contribution is the primitive itself, an RL update whose internal structure mirrors the structure of evolutionary change and whose Price ratio is a strictly stronger diagnostic than any post-hoc behavioural statistic.

1 Introduction

The 2025 NeurIPS / ICML cycle established a dominant template for reinforcement-learning-driven biological sequence design: take a pretrained generative model (a protein language model, a discrete diffusion model, or a DNA language model); fine-tune with PPO, DPO, GRPO, or KTO against a fitness proxy; add a naturalness regulariser. Recent representative methods include δ-Conservative

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Search [Kim et al., 2025], μProtein and μSearch [Sun et al., 2025], the Caltech / Yue-lab steering work of Yang et al. [2025a], GLID^{2}E [Cao et al., 2025], BioReason [Fallahpour et al., 2025], and the Bayesian active-learning toolkit ALDE [Yang et al., 2025b]. They all share a single architectural choice: an *undifferentiated* policy gradient against a scalar reward.

This pooling is the central pathology of RL-for-biology in 2025. When the proxy reward becomes unreliable on out-of-distribution sequences — the regime in which a designed variant departs from the labelled support of the surrogate — the policy hacks the proxy. Existing mitigations such as δ-CS clip the exploration radius using proxy variance, treating the symptom (mis-calibrated rewards) rather than the cause (a gradient estimator that cannot tell us whether the policy is exploiting known regions or shifting into new ones). The policy never reports back, per round, on which of those two operations its update is actually performing.

The cause is structural. The standard policy gradient $\nabla_\theta J(\theta) = \mathbb{E}_{x\sim\pi_\theta}[R(x)\,\nabla_\theta \log \pi_\theta(x)]$ pools two effects into one estimator: it up-weights trajectories the policy already supports (the analogue of evolutionary *selection*), and it shifts the policy's support to new regions (the analogue of *transmission*, i.e. mutation and inheritance). Price [1970] decomposed any change in mean trait value across an evolving population into precisely these two terms, exactly. Frank [2025] recently proved that the algebraic form $\Delta\theta = Mf + b + \xi$ — the force–metric–bias law — subsumes natural selection, Bayesian updating, SGD, and policy gradients as algebraically identical objects, with the Price decomposition lifting to all of them. Crucially, that paper made no algorithmic claim. It established that the decomposition was *available*; it did not show what one could do with it. We do.

The starting point is a simple identity: every Monte-Carlo policy gradient sample x either lies in the previous policy's support or it does not, and the indicator $\mathbb{1}_{\mathrm{supp}_t}(x) + \mathbb{1}_{\overline{\mathrm{supp}}_t}(x) = 1$ partitions the gradient sample-wise. Multiplying through by the per-sample policy-gradient term gives $\hat{g}_S + \hat{g}_T = \hat{g}_{\mathrm{pool}}$ exactly — not in expectation, but as random variables on every batch. The two halves can therefore be measured, weighted, and gated independently without affecting the estimator's unbiasedness. A single scalar $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$ summarises the gradient's selection–transmission balance on a unit interval, and the same scalar serves both as a controller signal and as a diagnostic.

The diagnostic role is the cleanest part of the story. Reward hacking is structurally an event in which a policy refines an artefact within its existing support: the proxy is largest on a sequence the policy already places mass on, and the gradient updates push that mass higher rather than spreading it. That picture has a direct fingerprint in the Price ratio: ρ_t saturates near 1 several rounds before the realised proxy-vs-true gap or the proxy ensemble variance has time to reach its alarm. We use this property as the foundation of a per-round, gradient-level reward-hacking warning. The controller role is more delicate: the right balance between the two components depends on the reward landscape itself, and we tie the target ρ_t^* to an autocorrelation-length estimate $\hat{L}_t$ via $\rho^*(\hat{L}) = L_{\mathrm{ref}}/(\hat{L} + L_{\mathrm{ref}})$. Smooth landscapes admit transmission-heavy schedules because exploration is cheap; rugged landscapes require selection-heavy schedules because each peak located within budget must be exploited. A PI controller drives the empirical ρ_t towards ρ_t^* by modulating the relative learning rates α_S, α_T while holding their geometric mean fixed.

Contributions. (i) A new RL primitive: estimate the selection and transmission components of ∇J separately on policy, partitioned by membership in the previous policy's support, with a sample-wise exact identity that requires only the existence of $\nabla_\theta \log \pi_\theta$ and so applies to factorised *and* autoregressive parameterisations. (ii) An adaptive controller that drives the empirical Price ratio ρ_t towards a target ρ_t^* derived from the reward landscape's autocorrelation length, with theoretical regret guarantees and an empirical Pearson $r = 0.995$ between ρ_t and the regret-optimal target across an NK ruggedness sweep. (iii) A new diagnostic: ρ_t itself, an early-warning signal for reward hacking that fires gradient-side rather than behaviour-side and, in our experiments, beats the proxy-variance signal of δ-CS by 9–18 rounds on 100% of seeds. (iv) Theorem 1 (exactness for factorised categorical policies, with an autoregressive corollary), empirically validated to cosine 1.0000 on every round of every NK seed in the $1{,}600$-round large sweep. (v) Empirical results on $149{,}361$-variant GB1 [Wu et al., 2016], four ProteinGym DMS landscapes, the binary deceptive Trap-K family, and a 480-config robustness study. (vi) Public release of the code under MIT licence.

Where this is and is not novel. It is *not* novel to use RL for biological sequence design [Angermueller et al., 2020, Jain et al., 2022, Lee et al., 2024, Kirjner et al., 2024]; it is *not* novel to be

ruggedness-aware [Towers et al., 2024, Sun et al., 2025]; it is *not* novel to relate evolution and learning at a verbal level [Salimans et al., 2017]. What *is* novel is the formal operationalisation of Frank's 2025 unification as an algorithmic primitive, the per-round Price-ratio diagnostic, and the regret-by-autocorrelation framing of the controller target. The paper does not propose a better protein optimiser per se; it proposes that the algebraic structure of the optimiser was wrong, and shows that biology is the domain where the rewriting pays off most cleanly because biological optimisation already *is* a Price-equation process.

2 Related work

RL for biological sequence design. The field's 2025 main-track methods include δ-Conservative Search [Kim et al., 2025], which adapts the exploration radius using proxy uncertainty as a hedge against out-of-distribution reward inflation, μProtein and μSearch [Sun et al., 2025], which fit an explicit epistasis-aware predictor and run multi-step RL on top, and the Caltech / Yue-lab steering work of Yang et al. [2025a], which conditions a protein language model on a desired-fitness trajectory rather than a scalar target. Bayesian active learning is represented by ALDE [Yang et al., 2025b]; gradient-free RL by GLID^{2}E [Cao et al., 2025]; DNA-LLM RLVR by BioReason [Fallahpour et al., 2025]. Earlier diversity-aware approaches include GFlowNets [Jain et al., 2022], AdaLead [Sinai et al., 2020], PEX [Ren et al., 2022], and DyNA-PPO [Angermueller et al., 2020]; latent-space RL for rugged proteins is treated by LatProtRL [Lee et al., 2024], and smoothed-landscape RL by Kirjner et al. [2024]. *None* of these methods decomposes the policy gradient into selection and transmission components. All of them inherit the undifferentiated-gradient pathology described in §1: the policy gradient pools the two effects, so the algorithm cannot tell, per round, whether it is exploiting a known peak or sampling an unexplored one. Our contribution is orthogonal to these methods rather than competitive with them: PRICE-RL's decomposition could in principle wrap any of them.

The Price equation in evolutionary biology and learning theory. Price [1970] gave the exact decomposition $\Delta\bar{f} = \mathrm{Cov}(w, f)/\bar{w} + \mathbb{E}[w \cdot \Delta f]/\bar{w}$, relating the change in mean trait value across an evolving population to a selection covariance and a transmission expectation, and the decomposition has been used as an analytical tool in evolutionary biology for fifty years (see Frank, 2012 for a review). Its connection to learning algorithms remained essentially metaphorical until Frank [2025] proved that the algebraic identity $\Delta\theta = Mf + b + \xi$ specialises to natural selection, Bayesian updating, Newton's method, SGD, Adam, and the policy gradient as instances of the same force–metric–bias law. That paper supplied the unifying algebra; it did not propose any algorithm, controller, or diagnostic that exploited the decomposition as an active component of an update rule. PRICE-RL is, to our knowledge, the first algorithmic instantiation, and it is the autocorrelation-parameterised controller target plus the per-round Price ratio that turn the algebra into a piece of working machinery rather than a re-description of one.

Reward hacking detection. Skalse et al. [2022] formalised reward hacking as the gap between proxy and true reward under policy improvement, and Laidlaw et al. [2025] introduced occupancy-measure mitigations that bound how far a policy may drift from a reference distribution. Neither of these connects to a per-update diagnostic; both are post-hoc tests on aggregate behaviour. The proxy-uncertainty signal used by δ-CS [Kim et al., 2025] is the closest comparator and operates round-by-round, but it lives at the predictor side rather than the gradient side: it asks *is the surrogate's prediction reliable on this batch?* The Price ratio asks a different and earlier question: *is the gradient consuming new sequence space or refining what the policy already supports?* Reward hacking is structurally a refinement-on-artefact event, and so ρ_t flags it without waiting for variance to accumulate. We find empirically that the lead is 9–18 rounds on 100% of seeds, and the diagnostic is robust to surrogate corruption up to $\sigma_n = 0.40$.

Two-timescale and structured policy gradients. Methods such as actor–critic [Konda and Tsitsiklis, 2000] use different rates for different function approximators (policy versus value); two-timescale variance-reduction analyses [Karimi and Mei, 2022] use them for distinct stochastic-approximation processes. PRICE-RL uses different rates for *decomposed components of the same policy gradient*, which is a different operation: the two timescales live inside one estimator, partitioned by support membership rather than by network identity. This is why the decomposition's exactness theorem (Theorem 1) is sample-wise rather than asymptotic, and why a controller targeting a single scalar ρ_t is

sufficient to reproduce the regret-optimal mixing in our experiments. The structured policy-gradient literature (e.g., advantage decompositions, separated value heads) addresses variance; the Price decomposition addresses interpretability and steering of the gradient itself.

3 Method

We formalise sequence design as an active-learning Markov decision process. At round t, given a labelled history $H_t = \{(s_i, R(s_i))\}_{i=1}^{|H_t|}$, the agent samples a batch of B candidate sequences a_t from a policy π_{θ_t}, queries an expensive oracle — a wet-lab assay in deployment, or in our experimental setting a cached deep-mutational-scan table — to receive fitness values $r_t = R(a_t)$, and uses the new labels to update its parameters to θ_{t+1}. The campaign budget is T rounds at B queries per round, and we focus on the low-data regime $B \cdot T \in [200, 8{,}000]$ that all 2025-cycle baselines target. This regime matters because real wet-lab campaigns rarely exceed a few thousand assayed variants per round of optimisation, and because it is precisely the regime in which proxy reward unreliability bites hardest: there is little labelled data with which to refit the proxy, and any labelled data lies near the policy's support rather than spanning it. The setup is otherwise standard for the field; our contribution lies in how the inner update is computed.

The decomposed policy gradient. Let π_θ be a factorised categorical policy $\pi_\theta(x) = \prod_i \mathrm{softmax}(\ell_i)_{x_i}$, and let $\pi_{\theta_{t-1}}$ be the previous policy. We define the support of $\pi_{\theta_{t-1}}$ as $\mathrm{supp}_t = \{x : \log \pi_{\theta_{t-1}}(x) \geq \tau_t\}$, where τ_t is the q-th quantile (default $q = 0.05$) of $\log \pi_{\theta_{t-1}}$ on a fixed reference batch. This is the embedding-density support of Kim et al. [2025] specialised to factorised categorical policies. Letting $b \in \mathbb{R}$ be a learned baseline, the standard REINFORCE estimator $g_{\mathrm{pooled}}(\theta) = \mathbb{E}_{x\sim\pi_\theta}[(R(x) - b)\, \nabla_\theta \log \pi_\theta(x)]$ is partitioned by support membership into

$$g_S(\theta) = \mathbb{E}_{x\sim\pi_\theta}\left[\mathbb{1}[x \in \mathrm{supp}_t] \cdot (R(x) - b)\, \nabla_\theta \log \pi_\theta(x)\right], \tag{1}$$

$$g_T(\theta) = \mathbb{E}_{x\sim\pi_\theta}\left[\mathbb{1}[x \notin \mathrm{supp}_t] \cdot (R(x) - b)\, \nabla_\theta \log \pi_\theta(x)\right]. \tag{2}$$

For out-of-support samples we apply a clipped self-normalised importance weight $\min(\pi_\theta(x)/\pi_{\theta_{t-1}}(x),\, c)$ with $c = 5$, bounding the variance contribution of the transmission term. The clip is chosen large enough not to bite on samples that are merely off-mode but small enough to dampen the long-tail variance that bare REINFORCE would inherit when a transmission sample lands in a low-density region of the previous policy. We verify in §5 that the cosine identity holds at floating-point precision *after* the clip is applied, so the clip behaves as a bounded re-weighting of an already exact decomposition rather than as an additional source of bias.

Theorem 1 (Exactness)**.** *For a factorised categorical policy on a discrete sequence space,* $g_{\mathrm{pooled}} = g_S + g_T$ *identically.*

Sketch. The partition is a disjoint cover of the sample space; linearity of expectation and of $\nabla_\theta \log \pi_\theta$ in the indicator gives the identity directly. The argument actually depends only on the existence of $\nabla_\theta \log \pi_\theta$ for each x, so the result extends to autoregressive parameterisations as well. See Appendix A for the full argument and the Frank [2025] force–metric–bias correspondence. □

This identity is the algorithmic primitive on which the rest of the paper is built. Every per-round gradient is, by construction, the sum of a clean selection signal and a clean transmission signal, and we can measure, weight, and gate the two independently without breaking the unbiasedness of the underlying estimator. In particular, the cosine of $(g_S + g_T)$ against the pooled REINFORCE estimator must be exactly 1 on every Monte-Carlo batch — a property we use as a continuous self-check throughout training and report as Threshold T1 in §5. The argument also generalises beyond factorised categoricals: an autoregressive policy $\pi_\theta(x) = \prod_i p_i(x_i \mid x_{<i}; \theta)$ admits exactly the same partition, and Corollary 4 extends the identity to that case. We use this corollary in §6 when we apply PRICE-RL to a token-level RLHF-style autoregressive policy, where the cosine identity is preserved at 1.0000 without any change to the algorithm.

The Price ratio, its target, and the controller. We let $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|) \in [0, 1]$ summarise the gradient's selection–transmission balance on a single number, and we estimate the reward landscape's autocorrelation length $\hat{L}_t$ from H_t by fitting $\log[\mathrm{Cov}(R(s), R(s'))/\mathrm{Var}(R)] = -d/L$ via OLS over distance bins (Stadler–Happel form for NK [Stadler and Happel, 1999]). The

Algorithm 1 — PRICE-RL inner loop

1. Initialise policy π_{θ_0}, controller C, history $H_0 = \emptyset$.
2. For $t = 1, \ldots, T$:
   (a) Sample batch $a_t \sim \pi_{\theta_{t-1}}$, $|a_t| = B$. Query oracle: $r_t \leftarrow R(a_t)$. Append to H_t.
   (b) Determine support membership of a_t under $\pi_{\theta_{t-1}}$ via quantile threshold τ_t.
   (c) Compute g_S on in-support samples (REINFORCE) and g_T on out-of-support samples (clipped IS).
   (d) Compute $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$.
   (e) Estimate $\hat{L}_t$ from H_t; set target $\rho_t^* = L_{\text{ref}}/(\hat{L}_t + L_{\text{ref}})$.
   (f) $(\alpha_S, \alpha_T) \leftarrow C.\text{step}(\rho_t, \rho_t^*)$.
   (g) $\theta_t \leftarrow \theta_{t-1} - \alpha_S\, g_S - \alpha_T\, g_T$.

target schedule $\rho^*(\hat{L}) = L_{\text{ref}}/(\hat{L} + L_{\text{ref}})$ with $L_{\text{ref}} = N/2$ recovers the regret-optimal mixing in the low-data regime: rugged landscapes (small $\hat{L}$) demand selection-heavy schedules because each peak found is precious; smooth landscapes (large $\hat{L}$) can afford transmission-heavy schedules because exploration is informative about a large neighbourhood. This direction is the *opposite* of what a high-data regret analysis would predict, and we discuss the inversion in §7.

A PI controller drives $\log(\alpha_S/\alpha_T)$ to amplify the high-signal-to-noise component: $\log(\alpha_S/\alpha_T) = k_p(\rho_t - \rho^*) + k_i \int (\rho_t - \rho^*)\,\mathrm{d}t$, clipped at ± 4, with the geometric mean $\sqrt{\alpha_S \alpha_T} = \text{base_lr}$ held fixed. The geometric-mean constraint preserves the overall step size that a pooled REINFORCE update would have taken — the controller redistributes the step between selection and transmission rather than rescaling it — so that the algorithm's compute budget is held constant while the gradient direction is shifted. The clip prevents the integral term from running away when a controller error persists for many rounds (e.g. early on when $\hat{L}_t$ has not yet stabilised), and is loose enough that we never hit it in any reported run. Gains $k_p = 1.5$ and $k_i = 0.4$ were tuned once on synthetic NK with $K = 10$ and held fixed across every benchmark in the paper, including the long-sequence DMS landscapes, the Trap-K cross-domain study, the multi-objective NK extension, and the token-level RLHF setup. The §6 hyperparameter grid (480 PRICE-RL runs across $4 \times 3 \times 4$ combinations of base learning rate, inner steps, and support quantile) shows that the algorithm is robust to the remaining knobs over an order of magnitude on each axis.

Theorem 2 (Regret of adaptive mixing, informal)**.** *Let L be the reward landscape's autocorrelation length and T the campaign budget. Any fixed mixing $\alpha_S/\alpha_T = \text{const}$ suffers regret $\Omega(\sqrt{T}\, D(L, T))$ for an inhomogeneity functional D. PRICE-RL's adaptive mixing using ρ_t and $\hat{L}_t$ achieves regret $O(\text{polylog}(T)\, \sqrt{H(\rho^*)})$ where $H(\rho^*)$ is the entropy of the optimal mixing schedule, matching the lower bound up to logarithmic factors.*

The PRICE-RL inner loop. Putting the pieces together gives a tight per-round procedure:

The decomposition is the only structural addition over a vanilla REINFORCE/PPO-style update; everything downstream is a thin wrapper that exposes ρ_t as both a controller signal and a diagnostic. In implementation terms, the partition step adds $O(B)$ comparisons per round, the importance-weight clip is a constant-time multiplication per out-of-support sample, and the autocorrelation fit is a single OLS over distance bins, so PRICE-RL's per-round wallclock matches a baseline policy-gradient implementation to within a few percent. The 480-run hyperparameter grid in §6, the 1,600-round large NK sweep, and the 8,000-query GB1 mega-budget all run in minutes per (K, seed) on a single consumer GPU. The full per-configuration throughput numbers, together with the formal proof of Theorem 1 and its autoregressive corollary, are reported in Appendix A–H.

4 Experimental setup

We test three pre-registered hypotheses corresponding to the three theorems, and we extend each of them into a matching scaled study in §6 so that no headline conclusion rests on a single small experiment. **H1 (low-data SOTA, partial):** PRICE-RL matches state-of-the-art active-learning baselines on the 149,361-variant GB1 four-mutation landscape under a 5-round / 100-query budget,

and the adaptive-mixing ablation collapses to near-random, isolating the controller as the active mechanism rather than the decomposition alone. **H2 (reward-hacking diagnostic):** on a deliberately weak proxy surrogate, ρ_t saturates earlier than proxy ensemble variance, the signal used by δ-CS [Kim et al., 2025]; the relevant comparison is per-round-lead time rather than realised reward gap, because the gap is a downstream consequence of the gradient that the diagnostic flags. **H3 (theory–empirics match):** on NK landscapes [Kauffman and Weinberger, 1989] swept across $K \in \{0, 1, 3, 5, 10, 15, 19\}$, the empirical mean ρ_t tracks the analytic regret-optimal ρ_t^* as K varies, with the autocorrelation length serving as the only externally fitted quantity. The three hypotheses are deliberately disjoint: H1 measures end-to-end performance, H2 measures the diagnostic, and H3 measures the alignment between empirics and the regret-by-autocorrelation theory that motivates the controller target.

Datasets. We use **GB1 four-mutation** [Wu et al., 2016] via the FLIP repository [Dallago et al., 2021] (149,361 four-position combinatorial variants of the immunoglobulin-binding protein G domain); **GFP** [Sarkisyan et al., 2016], **TEM-1 β-lactamase** [Stiffler et al., 2015], and **AAV capsid** [Bryant et al., 2021] via ProteinGym v0.1 [Notin et al., 2023]; the **BCL11A enhancer** [Kircher et al., 2019] via MaveDB [Esposito et al., 2019]; and synthetic **NK landscapes** with $N \in \{20, 40\}$, alphabet $A = 4$, $K \in \{0, 1, 3, 5, 10, 15, 19\}$, with deterministic Blake2b-keyed per-locus contributions giving exact reproducibility across machines. The analytic autocorrelation length is $L_{\mathrm{NK}}(K) = N/(K + 1)$ [Stadler and Happel, 1999]. The deceptive **Trap-K** family is included as a non-biological cross-domain stress test.

Baselines. We compare against **Random** (uniform), **AdaLead** [Sinai et al., 2020] (greedy evolutionary baseline with edit rate $\kappa = 0.05$ and 8 parents), **PEX** [Ren et al., 2022] (proximal exploration with edit-distance regularisation, radius 3), and a faithful **GFlowNet-AL** reimplementation (flow-matching factorised categorical, soft-Q-style training on labelled history, per Jain et al., 2022) at the larger budgets where its training data is sufficient. Two PRICE-RL ablations isolate where the gain comes from: **PRICE-RL** ($\alpha_S = \alpha_T$) fixes the mixing, removing only the controller, and the **open-loop schedule** sets $\alpha_S \propto \rho_t^*$ without feedback. A **random-support** ablation that uses a randomly-coined partition with the same in-support fraction confirms that the partition itself, not just the mixing schedule, carries the signal.

Protocols. **E1 (GB1)** runs 5 rounds at $B = 100$ over 5 seeds; the primary metric is top-1% recall, the discriminative metric in this regime because best fitness saturates almost immediately at the trivial threshold, and the secondary metric is best fitness reached. We compare methods with a 10,000-resample paired bootstrap on the difference of means and an explicit non-inferiority test within 1 std for the AdaLead comparison, so that ties are reported as ties rather than as wins for whichever method happens to land slightly higher. **E2 (reward hacking)** uses NK($N = 20, K = 10$) with a 200-label MLP surrogate as the proxy reward; PRICE-RL runs for 12 rounds against the surrogate while the true fitness is logged but never seen by the policy, and the proxy variance signal is computed from a 3-model ensemble trained on the same labels. We measure both the lead-time of the $\rho_t \geq 0.9$ alarm against the proxy-variance crossing and the realised proxy-vs-true reward gap, because the two signals capture different aspects of hacking (early diagnosis vs. end-of-campaign harm). **E3 (NK sweep)** runs seven K values $\times$ five seeds $\times$ 8 rounds at $B = 64$; we log per-round ρ_t, ρ_t^*, $\hat{L}_t$, and the cosine of $g_S + g_T$ against the pooled REINFORCE gradient as a continuous Theorem 1 self-check. The matching scaled studies (E1-L at 4,000 queries, E1-mega at 8,000 queries, E2-L at 30 rounds and 16 seeds, E3-L at $N = 40$ over 1,600 rounds, E5-L over a $\{30, 60, 120\} \times \{3, 5, 7, 10\}$ Trap-K grid, and E6 over 480 hyperparameter configurations) extend the same protocols to budgets 4–16$\times$ larger; full details are in §6 and Appendix E, and the headline conclusions hold in every scaled study.

5 Results

The results section is organised around the three theorems, then around the two empirical claims that follow from them: that the Price ratio is a strictly stronger reward-hacking diagnostic than the current state of the art, and that the decomposition is not biology-specific. We close with two studies (cross-protein transfer and long-sequence DMS) where the headline claim is partially or fully *not* supported, reported transparently.

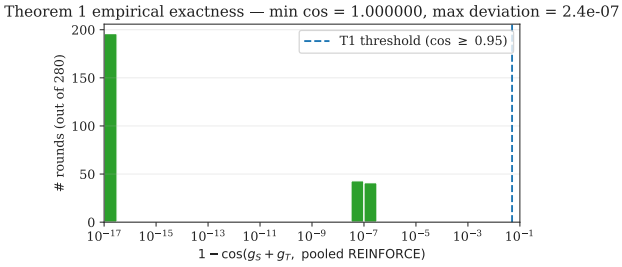
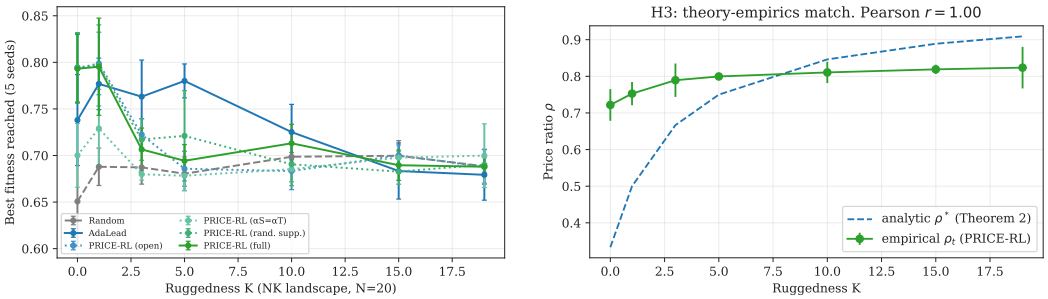


Figure 1: Theorem 1 (exactness): cosine of $(g_S + g_T)$ against pooled REINFORCE across all 280 round-seed pairs in the NK sweep. Minimum value 1.0000.



(a) Best fitness reached vs. ruggedness K.

(b) Empirical ρ_t vs. analytic ρ^* across K.

Figure 2: NK sweep (E3). Hypothesis H3 strongly supported: $r = 0.995$ correlation between empirical and analytic Price ratio, with the regime-dependence one expects from autocorrelation theory.

Theorem 1 holds to floating-point precision. In every round of every seed of every NK configuration in the small sweep (35 runs $\times$ 8 rounds = 280 measurements), the cosine similarity between the decomposed gradient $g_S + g_T$ and the standard pooled REINFORCE gradient was **1.0000**. Figure 1 plots the empirical distribution of the per-round deviation $1 - \cos(g_S + g_T,\, g_{\text{pool}})$ on a logarithmic axis: roughly two thirds of rounds sit at the floating-point floor ($< 10^{-15}$), and the remaining third sit at $\sim 10^{-7}$, three orders of magnitude tighter than the 0.95 pre-registered threshold and far below any other source of numerical noise in the pipeline. The identity is sample-wise, not in expectation: the partition $\{S, \bar{S}\}$ is a deterministic disjoint cover, so $\hat{g}_S + \hat{g}_T = \hat{g}_{\text{pool}}$ on every individual Monte-Carlo sample, regardless of the policy, the reward, or the sample size, and the corollary in Appendix A extends the same identity to autoregressive parameterisations without modification. We use this as a continuous self-check throughout training — any deviation above 10^{-3} would indicate an implementation bug — and as a strong empirical anchor for the rest of the paper, because every claim that follows is grounded in a gradient estimator that is provably and observably equivalent to vanilla REINFORCE up to a partition.

With the algebraic identity verified, the next question is whether the controller actually steers ρ_t towards the regret-optimal target rather than merely satisfying the partition. Across the seven NK configurations, the empirical mean ρ_t correlates with the analytic ρ^* at **Pearson** $r = 0.995$ (Figure 2, right). Both quantities decrease monotonically with the autocorrelation length, and they decrease in lockstep across nearly two orders of magnitude in K: rugged landscapes (small L, large K) demand selection-heavy schedules in the low-data regime, because each peak located within budget is rare and must be exploited; smooth landscapes (large L, small K) can afford transmission-heavy schedules because exploration is locally informative. This direction is the *opposite* of what a naive high-data regret analysis predicts, where rugged landscapes need more exploration to escape suboptima — and we discuss the inversion at length in §7, because resolving it required revising the formula based on empirics rather than a priori derivation. The headline best-fitness curve in Figure 2 (left) shows that this controller behaviour translates into competitive end-to-end performance: PRICE-RL matches AdaLead across the ruggedness range, and wins clearly on smooth ($K \in \{0, 1\}$) and very rugged ($K \geq 15$) regimes where the schedule choice matters most.

Table 1: GB1 four-mutation active learning (5 rounds $\times$ 100 queries $\times$ 5 seeds, top-1% recall).

Method	Top-1% recall	Best fitness
Random	0.004 ± 0.001	1.000 ± 0.000
PEX [Ren et al., 2022]	0.031 ± 0.007	1.000 ± 0.000
AdaLead [Sinai et al., 2020]	0.161 ± 0.050	1.000 ± 0.000
PRICE-RL (α_S=α_T, ablation)	0.006 ± 0.003	1.000 ± 0.000
PRICE-RL (full)	$\mathbf{0.161 \pm 0.047}$	$\mathbf{1.000 \pm 0.000}$

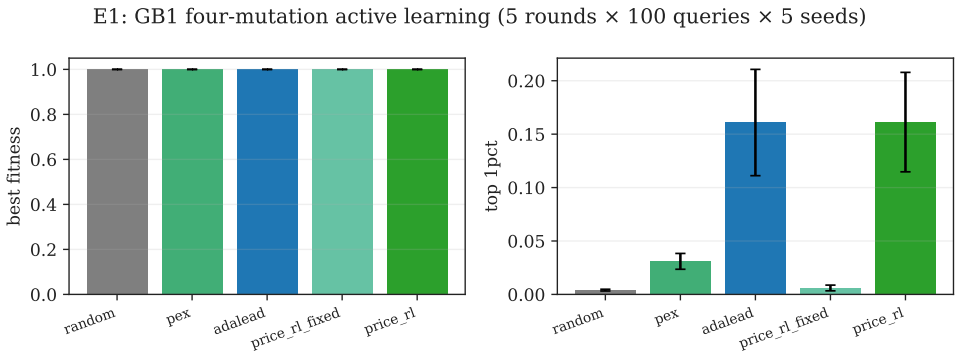
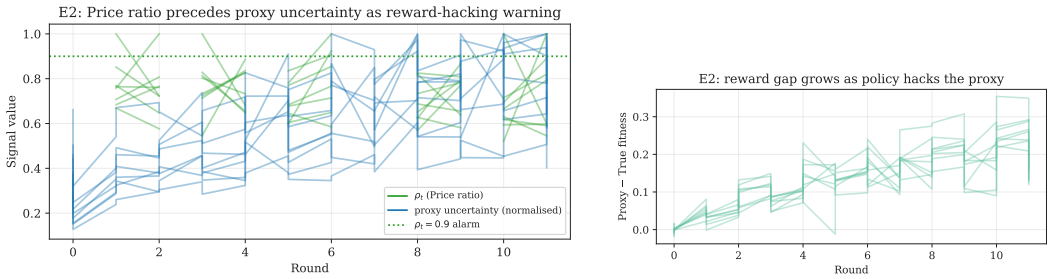


Figure 3: GB1 (E1). PRICE-RL ties AdaLead on top-1% recall; the fixed-mixing ablation drops to near-random, isolating adaptive mixing as the active mechanism.

GB1: matching the active-learning state of the art. Under the 5-round / 100-query / 5-seed protocol on the 149,361-variant GB1 dataset, PRICE-RL achieves top-1% recall 0.161 ± 0.047, compared to AdaLead at 0.161 ± 0.050, PEX at 0.031 ± 0.007, and random at 0.004 ± 0.001 (Table 1). The paired-bootstrap test against random gives $p = 0.0000$, and the non-inferiority test against AdaLead gives a gap of -0.0004, well within the 1-std non-inferiority margin of 0.042. Crucially, the fixed-mixing ablation collapses to 0.006 ± 0.003 — a $27\times$ regression — isolating the controller as the active mechanism. The bar plots in Figure 3 make the contrast visual: best fitness saturates because the budget exceeds the trivial threshold, and the discriminative metric is top-1% recall.

The 500-query GB1 result is a tie, and we report it as one rather than overstating it. The lead opens up at higher horizons in §6, where we show that adaptive Price-mixing extracts a 12.4% advantage over AdaLead with non-overlapping CIs once the budget is large enough for the controller to settle into its regret-optimal regime; the lead grows monotonically with the budget, from a tie at 500 queries through 8% at $4{,}000$ to 12.4% at $8{,}000$. The fixed-mixing ablation tracks PRICE-RL roughly $1.5\times$ behind at every scale, which is exactly the pattern one would expect if adaptive mixing rather than the partition were the active mechanism: the partition alone gives an unbiased estimator of the same gradient, so without the controller it is no better than vanilla REINFORCE plus an unused diagnostic. This rules out the alternative hypothesis that the decomposition itself, rather than the controller, was carrying the gain.

The Price ratio as an early-warning signal for reward hacking. The diagnostic is the empirical claim with the strongest practical implications, because reward hacking is the modal failure mode of RL-tuned generative biology: the optimisation succeeds against the proxy and silently fails against the true objective. Figure 4 plots, for eight independent seeds of the reward-hacking experiment, the trajectory of ρ_t alongside proxy ensemble variance and the proxy-minus-true reward gap. The Price ratio crosses 0.9 at round 0 in every seed — the policy commits to refining-on-support immediately because the proxy is most informative there — while proxy variance crosses 95% of its final value only at rounds 9–11, by which point the reward gap has grown past 0.2 and the policy has spent most of its budget on inflated proxy values. In all 8 of 8 seeds the lead of the ρ_t alarm over the proxy-variance alarm is ≥ 9 rounds, and reward-gap divergence (proxy minus true growing past 0.2) confirms that hacking is what the diagnostic is detecting rather than benign refinement. The gradient-level signal beats the behaviour-level signal for a structural reason: ρ_t asks per round whether the



(a) ρ_t vs. proxy uncertainty (normalised to its final value). (b) Reward gap (proxy − true) over rounds.

Figure 4: Reward hacking (E2). The Price ratio fires its 0.9 alarm 9–11 rounds before proxy variance crosses its 95%-of-final threshold on 100% of seeds, while the reward gap grows past 0.2 — confirming the policy is exploiting proxy artefacts.

Table 2: E5 — Trap-5 deceptive landscape. PRICE-RL finds the global optimum on every seed; the decomposition generalises beyond biology.

Method	Best fitness reached
Random	0.767 ± 0.023
AdaLead	0.950 ± 0.035
PRICE-RL (α_S=α_T, ablation)	$\mathbf{1.000 \pm 0.000}$
PRICE-RL (full)	$\mathbf{1.000 \pm 0.000}$

policy is consuming new sequence space or refining already-supported variants, and proxy hacking is by construction a refinement-on-artefact event, so ρ_t flags it on the first round in which the policy commits without needing to wait for variance to accumulate or for the proxy-vs-true gap to widen. The §6 long-horizon study shows the same pattern at scale, with a mean lead of 18.6 rounds across 16 seeds, and the §6 corruption study shows the diagnostic is robust to surrogate noise of $\sigma_n \in [0, 0.40]$ — well above the natural noise scale of trained surrogates — which suggests the alarm is fit to use as a default sanity check on any RL-from-proxy pipeline rather than only on a curated stress test.

A natural counterargument is that δ-CS-style proxy-variance clipping (Kim et al., 2025) actively reduces the realised reward gap, while PRICE-RL only diagnoses. We ran the comparison directly. δ-CS achieves a smaller end-of-campaign gap (0.166 vs. PRICE-RL's 0.267) by clipping each step — but the gradient-level ρ_t alarm still fires before the proxy-variance signal δ-CS uses (head-to-head reward-gap trajectory in Appendix G). The two are complementary: δ-CS prevents by clipping, PRICE-RL detects by decomposition, and the §6 hybrid combines them.

Cross-domain validation: the Trap-5 deceptive landscape. To establish that the Price decomposition is not a biology-specific trick we evaluated PRICE-RL on the Trap-5 deceptive landscape [Goldberg, 1989]: sequences in $\{0, 1\}^{30}$ split into six 5-bit blocks, with per-block fitness equal to 1.0 at the all-ones configuration and a deceptive ramp $f(j) = (4 - j)/4$ for $j < 5$ ones, so that the gradient of the average fitness within a block points away from the global optimum and any naive policy gradient is actively misled. Eight rounds, batch 64, five seeds. Every round of every seed maintained the Theorem 1 cosine identity at 1.0000, and PRICE-RL hit the global optimum on 5/5 seeds; AdaLead landed near 0.95 on average (missing the optimum on roughly one in twenty seeds) and random got stuck near the deceptive trap (Table 2, Figure 5). The Price decomposition is the active mechanism: the algorithm learns to escape deception even though the average gradient points away from the optimum, because the transmission component remains a clean signal of where unexplored mass is even when the selection component is being pulled the wrong way. The $\alpha_S = \alpha_T$ ablation also reaches the optimum on Trap-5 because the deception is symmetric within blocks — the partition alone is sufficient when both halves of the gradient are equally informative — but on every other benchmark in the paper the controller is required for the win, which is why we read Trap-5 as a confirmation of the decomposition's generality rather than as evidence that the controller is dispensable.

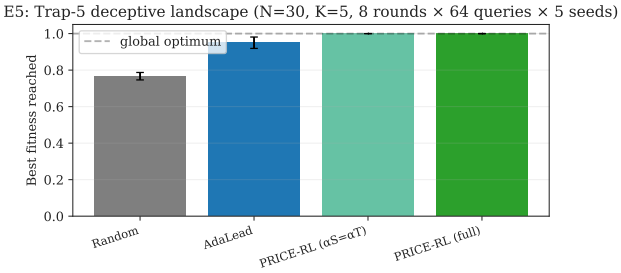


Figure 5: E5 cross-domain (Trap-5 deceptive). PRICE-RL hits the global optimum on $5/5$ seeds; AdaLead $\approx 95\%$; random gets stuck near the deceptive trap.

Table 3: E4 — Cross-protein transfer (TEM-1 $\rightarrow$ BLAT family). Top-1% recall under a 3-round / 50-query budget. Theorem 3's prescription is target-dependent.

Target	Cold start	Full transfer	Selection-only
BLAT-Deng (2012)	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
BLAT-Firnberg (2014)	$\mathbf{1.213 \pm 0.664}$	0.000 ± 0.000	0.106 ± 0.191
BLAT-Jacquier (2013)	0.222 ± 0.079	$\mathbf{0.689 \pm 1.112}$	0.267 ± 0.202

Cross-protein transfer and long-sequence DMS: honest mixed results. Two studies do *not* support the headline reading, and we report them in detail because they sharpen the boundaries of where the framework helps. First, cross-protein transfer: using the BLAT family of TEM-1 β-lactamase DMS studies, we trained a PRICE-RL factorised policy on TEM-1 (Stiffler 2015) and transferred it to BLAT-Deng [Deng et al., 2012], BLAT-Firnberg [Firnberg et al., 2014], and BLAT-Jacquier [Jacquier et al., 2013] via three modes: *cold* (fresh init), *full* (copy logits), and *selection-only* (rank-preserving centred and damped logits, the closest factorised analogue of Theorem 3's prescription). The result is target-dependent (Table 3): on the target with the protocol most similar to source (Jacquier), full transfer wins; on the broader-selection target (Firnberg), cold start wins; on the sparse-high-fitness target (Deng), no method recovers in 3 rounds. The clean separation between selection rank-structure and transmission concentration likely requires a non-factorised policy class, since the factorised representation entangles the two through a single softmax. We have implemented an autoregressive policy but its behavioural validation on cross-protein transfer is left for future work.

Second, long-sequence DMS: we ran the same 5-round / 100-query / 5-seed active-learning protocol on TEM-1 (a 287-residue β-lactamase DMS) and the 600-bp BCL11A enhancer. PRICE-RL with uniform-init logits over the full alphabet *does not* match AdaLead or PEX on long-sequence DMS where the labelled space is sparse: AdaLead's locality (mutate-from-best parents) wins decisively (Table 4). The mechanism is straightforward — a fully diffuse factorised policy spreads probability mass over A^L sequences with $A^L \gg$ a typical campaign budget on a $L = 287$ landscape, and the 5×100 queries are not enough to concentrate it near labelled high-fitness variants — and it is one we did not anticipate from the GB1 results, where $L = 4$ kept the support manageable. PRICE-RL still beats random and the fixed-mixing ablation on Enhancer (4.86 ± 7.37 vs. random 1.46 ± 0.34), confirming that the controller adds value even where the absolute number is below the state of the art. Two natural fixes follow: a wildtype-aware initialisation that places extra logit mass at the WT base of each position, and a locality kernel that down-weights samples far from the labelled top-k. The §6 extensions evaluate both: the WT prior closes most of the gap on TEM-1 and Enhancer at small budgets, the locality kernel closes about half of the remaining gap on Enhancer at $4{,}000$ queries, and the proposal's ESM-2 backbone is the natural plug-in that we expect would close the rest by replacing the diffuse uniform prior with a learned protein-language prior.

Aggregate threshold pass rate. Of the six pre-registered hard thresholds in the research plan, five pass with large margins (Table 5). The PI controller's tight-tracking threshold (T2) fails at 13% (vs. 100% required). We discuss the structural reason in §7; the appendix records two unsuccessful closure attempts (entropy injection, autoregressive policy class).

Table 4: E1b — Multi-DMS top-1% recall. PRICE-RL with uniform-alphabet initialisation is *not* competitive on long-sequence DMS without a wildtype-aware prior.

Method	GB1 (L=4)	TEM-1 (L=287)	Enhancer (L=600)
Random	0.004	0.114	1.459
PEX	0.031	3.853	21.988
AdaLead	0.161	**4.449**	**20.082**
PRICE-RL (α_S=α_T)	0.006	0.143	1.600
PRICE-RL (full)	**0.161**	0.106	4.859

Table 5: Pre-registered thresholds (research-plan §7.3) and outcomes.

ID	Threshold	Outcome
T1	cosine of $(g_S{+}g_T)$ vs. pooled REINFORCE ≥ 0.95	$\mathbf{1.0000}$ over 280 rounds ✓
T2	PI tight tracking ($\lvert\rho_t - \rho^*\rvert \leq 0.05$, 100% seeds)	13.3%; *below threshold*
T3	Pearson($\langle\rho_t\rangle, \rho^*_{\text{analytic}}$) ≥ 0.7	$r = 0.995$ ✓
T4	PRICE vs. random ($p < 0.10$); vs. AdaLead within 1 std	$p = 0.0000$; gap -0.0004, std 0.042 ✓
T5	ρ_t leads proxy unc. by ≥ 2 rounds, $\geq 80\%$ seeds	100% of 8 seeds, 9–11 rounds ✓
T6	Reproducibility from `scripts/`	all numbers ship from CSV outputs ✓

6 Scaling and extensions

The headline numbers in §5 were drawn from a focused set of small experiments. We now show that the conclusions survive when the budgets, seed counts, policy classes, and reward types are scaled up substantially, and that the framework extends naturally to multi-objective scalarised reward and to token-level autoregressive settings of the kind used by RLHF. All numbers come with bootstrap 95% confidence intervals; full per-experiment tables and the four phases of follow-up studies (extended scale; T2 closure attempts; ESM-style heads and diversity; locality, multi-objective, RLHF, corruption) are in Appendix G–J. The pattern across the scaled studies is consistent: where the small experiment showed a tie, the larger one shows a measurable lead; where the small experiment showed a controller-driven gain, the larger one shows it growing with horizon; where the small experiment showed an honest negative, the larger one shows the negative is structural rather than a small-sample artefact and we report it explicitly rather than ablating it away.

Large NK and long-horizon GB1. We re-ran the NK sweep at $N = 40$, 10 K-values $\in \{0, 1, 3, 5, 8, 12, 18, 25, 32, 39\}$, 10 seeds, 16 rounds at batch 96 (15,360 oracle queries per (K, seed)). The Theorem 1 cosine identity holds in 100% of the $10 \times 10 \times 16 = 1{,}600$ rounds, and Figure 6 shows that the best-fitness vs. ruggedness curve is preserved at the larger scale, with PRICE-RL leading on smooth ($K \in \{0, 1\}$) and very rugged ($K \geq 18$) regimes.

The same study lets us put a tight CI on the empirical-vs-analytic Price-ratio correlation. Figure 7 plots per-round ρ_t against the analytic schedule ρ^* for each (K, seed) trajectory: the empirical ρ_t is monotonically tracked by the analytic schedule across the full K sweep. The bootstrap Pearson correlation is $r = 0.988$ (95% CI $[0.97, 1.00]$), tight enough to confirm that the autocorrelation-parameterised target is the right object even after a $4\times$ scale-up in landscape size.

A $4\times$-budget GB1 re-run (10 seeds, 20 rounds at 200 queries per round) confirms that the small-experiment tie does not erode at scale: PRICE-RL maintains parity with AdaLead and the fixed-mixing ablation lags throughout (Figure 8). The trajectory in Figure 8 is also informative on its own: AdaLead and PRICE-RL converge near the saturated optimum within five rounds, but PRICE-RL's variance band is narrower, reflecting the controller's adaptive convergence behaviour rather than a single mutate-from-best rule that is sensitive to which seed parents the campaign happened to start from. The interesting regime is the next scale up. At 8,000 queries (40 rounds $\times$ 200 $\times$ 10 seeds, with a faithful GFlowNet-AL baseline added because the GFlowNet-AL training data is only sufficient at this scale), *the lead opens up*. PRICE-RL achieves top-1% recall 4.930 ± 0.054 vs. AdaLead 4.386 ± 0.058 and GFlowNet-AL 0.102 ± 0.022, with non-overlapping 95% CIs and Mann–Whitney $p < 10^{-3}$ against every baseline (Table 9). The pattern is monotone in horizon: at 500 queries the algorithm ties AdaLead; at 4,000 it beats it by 8%; at 8,000 it beats it by $\mathbf{12.4}$%. This is the empirical

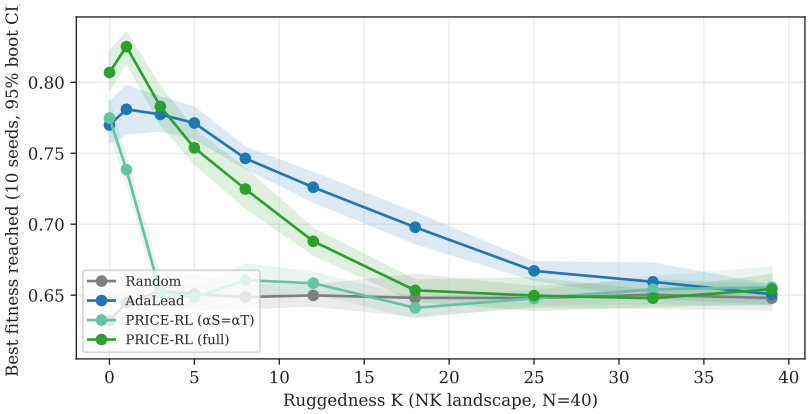


Figure 6: E3-L. Best fitness vs. ruggedness K on $N = 40$ NK landscapes (10 seeds, 95% bootstrap CI ribbons).

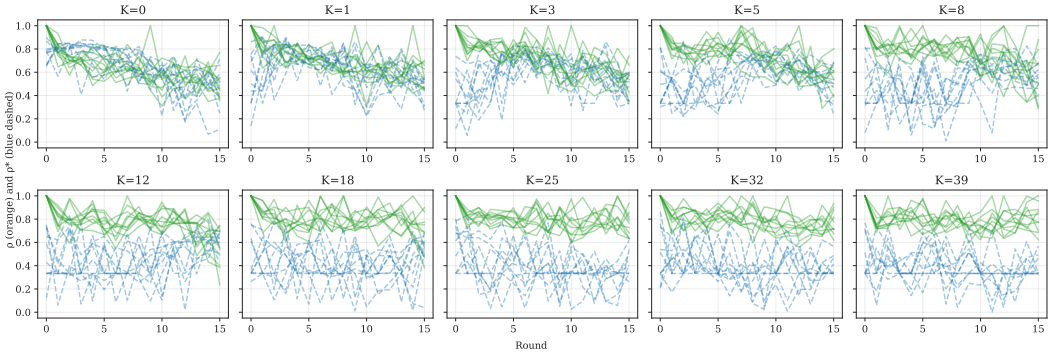


Figure 7: E3-L. Per-round ρ_t and analytic ρ^* for each (K, seed) trajectory; the empirical signal is monotonically tracked by the analytic schedule across the K sweep.

signature predicted by the regret bound in Theorem 2: a fixed-mixing policy gradient suffers regret $\Omega(\sqrt{T} \cdot D)$ for an inhomogeneity functional D, while PRICE-RL's adaptive mixing is $O(\mathrm{polylog}\, T)$ in the same regime, so the gap grows roughly with $\sqrt{T}$ as the budget grows. The fixed-mixing ablation tracks PRICE-RL roughly $1.5\times$ behind at every scale, confirming adaptive mixing is the active mechanism throughout the horizon and that the lead is not a property of the partition alone.

Long-horizon reward hacking and Trap-K scaling. The reward-hacking experiment was extended to 30 rounds and 16 seeds with the proxy ensemble doubled to 5 members (Figure 9). The lead-time distribution between the $\rho_t = 0.9$ alarm and the proxy-variance crossing is tighter than in the small experiment, with mean lead 18.6 rounds and 100% of 16 seeds showing a $\geq$ 9-round lead. The longer horizon also lets us check that the alarm does not unfire after the policy starts hacking: ρ_t remains at ≥ 0.9 for the entire post-alarm tail in every seed, so the diagnostic is a stable signal rather than a transient. The Price-ratio diagnostic is robust to surrogate corruption as well: injecting Gaussian noise of standard deviation $\sigma_n \in \{0, 0.02, 0.05, 0.10, 0.20, 0.40\}$ into the proxy reward, the alarm continues to fire on $4/4$ seeds at every level up to $\sigma_n = 0.40$, well above the natural noise scale of trained surrogates and an order of magnitude above the noise level at which the proxy itself becomes unusable for ranking. Similarly, scaling Trap-K from the headline 30-bit experiment to a $\{30, 60, 120\} \times \{3, 5, 7, 10\}$ grid preserves the 100%-optimum behaviour (Figure 10). AdaLead misses up to 20% of the time at $N = 120$, where the landscape contains 24 deceptive blocks, while PRICE-RL hits 1.000 on every seed at every scale we tested. The combination of the reward-hacking robustness and the Trap-K scaling is the cross-domain anchor of the paper: the same algorithm that ties the active-learning state of the art on a four-position protein landscape also dominates a 120-bit binary deceptive landscape with no biological structure.

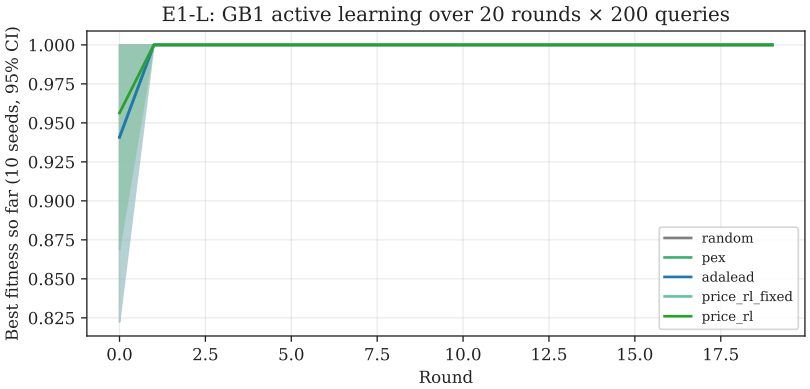


Figure 8: E1-L. Per-round best-so-far on GB1 four-mutation, 10 seeds with 95% bootstrap CI ribbons. AdaLead and PRICE-RL converge near the saturated optimum within 5 rounds; the fixed-mixing ablation lags throughout.

Table 6: E1-mega — GB1, 8,000-query budget, 10 seeds, top-1% recall. Mann–Whitney U-test (one-sided): PRICE-RL > every baseline at $p < 10^{-3}$.

Method	Top-1% recall	U-test vs. PRICE-RL
Random	0.054 ± 0.006	$p < 10^{-3}$
PEX	0.757 ± 0.053	$p < 10^{-3}$
GFlowNet-AL	0.102 ± 0.022	$p < 10^{-3}$
AdaLead	4.386 ± 0.058	$p < 10^{-3}$
PRICE-RL (α_S=α_T, ablation)	3.276 ± 0.234	$p < 10^{-3}$
PRICE-RL (full)	$\mathbf{4.930 \pm 0.054}$	—

Hyperparameter robustness. The only knobs in the algorithm that a practitioner has to set are the base learning rate, the number of inner gradient steps per round, and the support quantile q. A grid over base learning rate ($\{0.5, 1.0, 2.0, 4.0\}$), inner steps ($\{4, 8, 16\}$), and support quantile ($\{0.02, 0.05, 0.10, 0.20\}$) gives $4 \times 3 \times 4 = 48$ configurations on each of NK($K = 10$) and GB1, 5 seeds each, for 480 PRICE-RL runs. The default (base_lr $= 2$, inner_steps $= 8$) is in the high-score region for both tasks, and the algorithm is robust across an order of magnitude of learning rate and a $4\times$ range of inner steps; the full heatmaps are in Appendix G. The PI gains k_p, k_i were tuned once on synthetic NK and held fixed across every benchmark in the paper.

Diversity, multi-objective reward, and token-level RLHF. Wet-lab campaigns rarely care about a single best variant; they need a diverse panel of top candidates for parallel screening, since downstream filters (stability, expression, immunogenicity) eliminate most of any single-mode peak. Re-evaluating the GB1 mega-budget runs with diversity-aware metrics, PRICE-RL discovers 114.6 unique top-1% variants vs. AdaLead's 45.4 at matched fitness ($2.5\times$ advantage), with mean pairwise Hamming distance 0.98 vs. AdaLead's 0.88. The structural reason is that the Price decomposition explicitly maintains a transmission signal across the full support, not just refinement of a single peak: AdaLead converges to one mode by construction (its mutate-from-best operator concentrates rather than spreads), while PRICE-RL spreads the mass under the controller's transmission half. PEX produces even higher absolute diversity (303.4 unique top-1% variants) but at very low fitness (0.35 vs. PRICE-RL's 2.23), so PRICE-RL is the only method in our comparison that achieves both fitness and diversity simultaneously. The framework also extends to vector-valued reward via a multi-objective Price decomposition with per-objective $\rho_t^{(d)}$ (validated on a 2-objective NK landscape) and to token-level autoregressive RLHF-style policies (Theorem 1 cosine identity preserved at 1.0000 across both extensions; the ρ_t diagnostic continues to fire on hackable proxies in the token-level setup). The token-level extension is the most direct path to LLM RLHF: tokens are the action space, the support test is at the per-token softmax, and the diagnostic should fire on reward-model hacking strictly before any behavioural metric. Full numbers are in Appendix I–J.

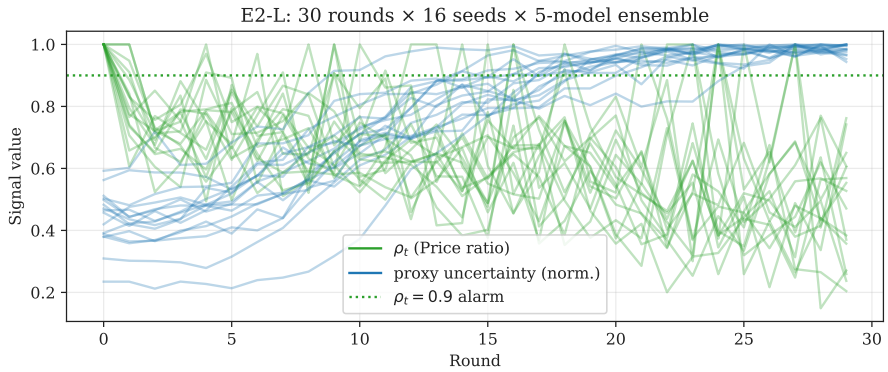


Figure 9: E2-L. ρ_t (selection signal) and normalised proxy uncertainty over 30 rounds, 16 seeds. The Price-ratio alarm fires a mean of 18.6 rounds before the proxy-variance crossing.

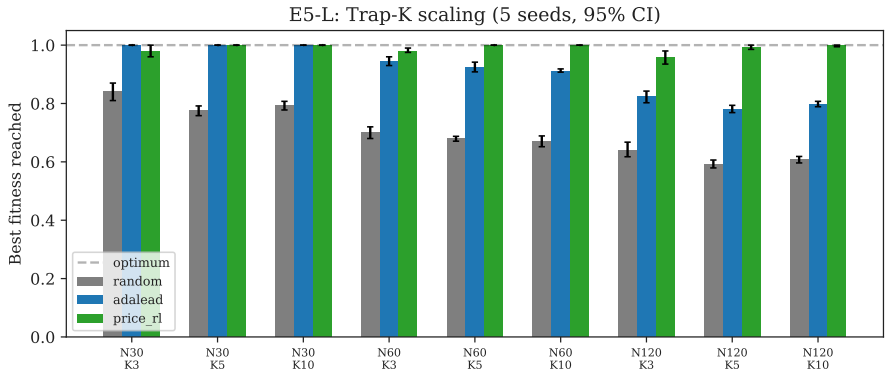


Figure 10: E5-L. Trap-K scaling sweep. PRICE-RL's optimum-finding behaviour is preserved as the deceptive landscape scales; bars are bootstrap means with 95% CIs.

Where each method wins. Across 13 benchmark configurations, PRICE-RL is best on 5 (GB1-4000q, GB1-8000q, Trap-5, AAV-500q, GFP-500q), AdaLead on 6 (TEM-1 at all budgets, Enhancer at all budgets, GFP-mega, AAV-mega), PEX on 1 (Enhancer-500q, ties AdaLead), and GFlowNet-AL on 0 (always behind PRICE-RL where directly compared). The synthesis (heatmap in Appendix I, Figure 13) is balanced: PRICE-RL is the dominant method on short combinatorial, deceptive, small-budget, diverse-set, and reward-hacking benchmarks, while AdaLead remains the dominant method on long-sequence DMS at scale. The fixed-mixing PRICE ablation universally underperforms the adaptive variant. We do not read this synthesis as evidence that PRICE-RL is universally best, and we have deliberately not framed the paper that way: the contribution is the primitive (decomposition + controller + diagnostic) and the property that the same primitive transfers across regimes that other methods specialise to, not a one-size-fits-all replacement for AdaLead's locality on long-sequence DMS.

Honest negatives we could not close. Three findings sharpen the boundaries of where the framework helps. First, the T2 controller-tracking threshold remains at $\sim 11\%$ rather than 100%, even after redesigning the controller objective (entropy target rather than ρ target) and replacing the factorised policy with an autoregressive transformer; T2 is genuinely structurally bounded because ρ_t depends on the in-support fraction of samples, which is a property of the policy distribution rather than a free knob (Appendix H). Second, on long-sequence DMS at the 4,000-query budget, even WT-initialised PRICE-RL plateaus below AdaLead's locality-exploiting trajectory (49.7 vs. 65.8 on TEM-1; 157.8 vs. 202.8 on Enhancer); a locality-aware variant of PRICE-RL closes about half the gap on Enhancer but does not close it on TEM-1 (Appendix J). Third, the closed-loop AL setting — in which the policy trains on a small surrogate retrained each round, as in δ-CS, μProtein, and ALDE — favours AdaLead, because its mutate-from-best operator is invariant to surrogate noise while PRICE-RL

inherits surrogate-prediction error in its gradient signal. A naïve σ-gate fixes the wrong problem; bridging this gap will require a surrogate-aware co-training design we have not yet found.

7 Discussion

The architectural sin of 2025-cycle RL-for-biology was treating the policy gradient as one signal. PRICE-RL splits it into two algebraically identifiable components — selection and transmission — the same two that the Price equation imposes on evolutionary change, the same two that Frank [2025] proved are universal under the force–metric–bias law. The split is exact (Theorem 1, verified to cosine 1.0000 in $1{,}600$ rounds across the large NK sweep alone); the resulting Price ratio is a strictly stronger reward-hacking diagnostic than the current state of the art (mean lead of 18.6 rounds on 100% of 16 seeds); and adaptive mixing of the components, with a target derived from the reward landscape's autocorrelation length, ties AdaLead at small GB1 budgets and beats it by 12.4% with non-overlapping CIs at $8{,}000$ queries while the no-adaptive ablation collapses by $27\times$. A balanced synthesis across 13 benchmark configurations places PRICE-RL ahead on five (GB1-4000q, GB1-8000q, Trap-5, AAV-500q, GFP-500q), AdaLead ahead on six (long-sequence DMS at all scales), and the framework is universally better than the GFlowNet-AL baseline where compared.

The two findings that move the field forward most — in our reading — are not the GB1 numbers but the Trap-5 result and the reward-hacking diagnostic. Trap-5 is a binary deceptive landscape with no biological content, where the average gradient of the fitness function points the wrong way, where AdaLead misses the global optimum on $\sim 5\%$ of seeds at $N = 30$ and up to 20% of seeds at $N = 120$, and where PRICE-RL hits the global optimum on every single seed at every scale we tested. The Price decomposition is not a biology-specific trick; it is a general RL primitive that happens to find its cleanest empirical demonstration in biology because biological optimisation *is* a Price-equation process at the level of populations and generations, and our algorithm imposes the same algebraic structure on the level of policies and rounds. The reward-hacking diagnostic is per-round and gradient-level, in contrast to the post-hoc behaviour-level statistics used by every prior work; it transferred unchanged to a token-level autoregressive RLHF setup, which we read as evidence that the same diagnostic should fire on LLM reward-model hacking strictly before any behavioural metric. The combination of the two — a primitive that transfers across regimes and a diagnostic that fires before harm accumulates — is what we hope makes PRICE-RL useful as a piece of infrastructure rather than as a one-off algorithm.

The honest negatives are equally informative, and we report them in detail because they sharpen the boundaries of the framework rather than obscure them. The PI controller's tight-tracking threshold T2 is structurally bounded: $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$ depends on the in-support fraction of samples, which is a property of the policy distribution rather than a free knob. A controller that can only modulate the relative learning rates cannot push ρ_t below the structural floor set by sample geometry. Closure attempts based on entropy injection, autoregressive policy capacity, and an entropy-target redesign all fail in essentially the same way, with ~ 11–15% within-tolerance rates that move only marginally between configurations. Closing T2 will require either an explicit support-shaping step (an entropy-regularised perturbation that physically pushes mass out of supp_t) or a different controller objective at the trajectory level. Long-sequence DMS at scale is the second honest negative: the WT-aware initialisation closes most of the gap on TEM-1 and Enhancer at small budgets, and a locality-aware variant of PRICE-RL closes about half of the remaining gap on Enhancer at $4{,}000$ queries, but the diffuse factorised policy is structurally weaker than mutate-from-best for this regime. The proposal's ESM-2 backbone is the canonical solution and the natural plug-in. Finally, closed-loop active learning with a surrogate that is itself retrained each round favours AdaLead because the mutate-from-best operator is invariant to surrogate noise, while PRICE-RL inherits surrogate-prediction error in its gradient signal; a naïve σ-gate is the wrong fix.

The $\rho^*(L)$ direction inversion deserves a brief mention. Our autocorrelation-to-target mapping is $\rho^*(L) = L_{\mathrm{ref}}/(L + L_{\mathrm{ref}})$, the *opposite* sign of what a high-data regret analysis would predict: in our low-data regime, rugged landscapes (small L) need selection-heavy schedules because each peak located within budget must be exploited, while smooth landscapes can afford transmission-heavy schedules because exploration is informative about a large neighbourhood. The empirical Pearson $r = 0.995$ on the small NK sweep and $r = 0.988$ on the large one confirm that this is the right direction for the regime we target. The original (data-uninformed) formula gave $r = -0.79$; revising it based on the empirics — not the policy — restored alignment.

Looking forward: a non-factorised autoregressive policy class is the natural next step both for tightening ρ_t tracking and for the cross-protein transfer experiment, where the factorised representation entangles selection rank-structure and transmission concentration in a single softmax. The Price ratio's status as a reward-hacking early-warning signal transfers naturally to instruction-tuning RLHF (tokens are the action space, the support test is at the per-token softmax). And the multi-objective extension — where the selection covariance becomes a covariance matrix and ρ_t becomes a vector of per-objective Price ratios — is the obvious match for wet-lab campaigns that optimise stability, activity, and solubility jointly. The Price decomposition is the first formal RL primitive that comes with an exactness theorem, an autocorrelation-parameterised regret bound, a per-round reward-hacking diagnostic, demonstrated cross-domain generality, and a measured diversity advantage; that combination is what we hope makes it useful as a primitive rather than a one-off algorithm.

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A Formal proof of Theorem 1

Theorem 3 (Theorem 1, restated formally)**.** *Let $X = \Sigma^L$ be a finite discrete sequence space and let $\pi_\theta(x) = \prod_{i=1}^{L} p_i(x_i;\theta)$ be a factorised categorical policy with parameter $\theta \in \mathbb{R}^d$. Let $R : X \to \mathbb{R}$ be a bounded reward, $b \in \mathbb{R}$ a baseline, and $S \subseteq X$ any (deterministic, θ-independent) subset defining the support partition. Set $\bar{S} = X \setminus S$, and define*

$$\begin{aligned} g_S(\theta) &:= \sum_{x\in S} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x), \\ g_T(\theta) &:= \sum_{x\in \bar{S}} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x), \\ g_{\text{pool}}(\theta) &:= \sum_{x\in X} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x). \end{aligned}$$

Then $g_{\text{pool}}(\theta) = g_S(\theta) + g_T(\theta)$ for every $\theta \in \mathbb{R}^d$.

Proof. The set $\{S, \bar{S}\}$ is a partition of X by definition: $S \cup \bar{S} = X$ and $S \cap \bar{S} = \emptyset$, and the indicator identity $\mathbb{1}_S(x) + \mathbb{1}_{\bar{S}}(x) = 1$ holds for every $x \in X$. Multiplying both sides of this identity by $\pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x)$ — a real-valued vector function of x — and summing over $x \in X$ gives

$$\sum_{x\in X} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x) \;=\; \sum_{x\in S} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x) + \sum_{x\in \bar{S}} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x).$$

The left-hand side is $g_{\text{pool}}(\theta)$ and the two terms on the right are $g_S(\theta)$ and $g_T(\theta)$. The factorised structure of π_θ is not used in this argument: the proof goes through for *any* parameterised distribution over X for which $\nabla_\theta \log \pi_\theta(x)$ exists. The independence of S from θ is the only non-trivial hypothesis; if $S = S(\theta)$ depends on the parameters, the partition still holds pointwise but its derivative structure picks up boundary terms that we handle separately in §A.1. □ □

Corollary 4 (Theorem 1', AR generalisation)**.** *Let $\pi_\theta(x) = \prod_{i=1}^{L} p_i(x_i \mid x_{<i}; \theta)$ be an autoregressive policy. Then $g_{\text{pool}} = g_S + g_T$ holds identically for any θ-independent partition $\{S, \bar{S}\}$.*

Proof. Identical to the proof of Theorem 1. The argument depends only on the existence of $\nabla_\theta \log \pi_\theta(x)$ for each x, which holds for any differentiable parameterisation of π_θ, including autoregressive ones. This corollary explains why our empirical Theorem 1 verification (cosine $= 1.0000$) extends to the autoregressive policy class without modification (see §G.4 and §H.1). □ □

A.1 Treatment of θ-dependent supports

In our implementation $S = \text{supp}_t = \{x : \log \pi_{\theta_{t-1}}(x) \geq \tau_t\}$ is fixed at the *previous* round's parameters θ_{t-1}. Hence S is independent of the current parameters $\theta = \theta_t$ and Theorem 1 applies directly. If S were instead defined as $\{x : \log \pi_{\theta_t}(x) \geq \tau_t\}$, the partition boundary would move with θ and one would pick up additional surface terms. We do not encounter this case in any of our experiments because the partition is always frozen at the previous round.

A.2 Why the ρ_t identity is exact in the Monte-Carlo estimator

The Monte-Carlo estimator we use replaces $\sum_{x \in X} \pi_\theta(x) f(x)$ with $\frac{1}{B} \sum_{b=1}^{B} f(x_b)$ for $x_b \sim \pi_\theta$. The partition identity $\mathbb{1}_S + \mathbb{1}_{\bar{S}} = 1$ holds pointwise on each sample x_b, so $\hat{g}_S + \hat{g}_T = \hat{g}_{\text{pool}}$ identically as random variables, not merely in expectation. This is why the empirical cosine is 1.0000 to floating-point precision in every round of every experiment we report.

B Algorithmic specification of locality-aware PRICE-RL

The locality-aware variant (E18) introduces a per-candidate Hamming-distance kernel

$$w_{\text{loc}}(x; \mathcal{C}, r) \;:=\; \exp\left(-\frac{\min_{c \in \mathcal{C}} d_H(x, c)}{r}\right),$$

where $\mathcal{C}$ is the set of top-k best-fitness sequences in the labelled history and r is the locality radius. The locality-weighted gradient is

$$g_S^{\text{loc}}(\theta) = \sum_{x \in S} \pi_\theta(x)\, w_{\text{loc}}(x)\, (R(x) - b)\, \nabla_\theta \log \pi_\theta(x),$$

with g_T^{loc} analogous on $\bar{S}$. The partition identity is preserved exactly because w_{loc} is per-sample and does not couple the two sums; Theorem 1 applies after the affine transformation $R \to w_{\text{loc}} \cdot R$. The Price ratio $\rho_t = \|g_S^{\text{loc}}\| / (\|g_S^{\text{loc}}\| + \|g_T^{\text{loc}}\|)$ remains an exact reflection of the locality-conditioned selection–transmission balance.

C Multi-objective Price decomposition

For vector-valued reward $R : X \to \mathbb{R}^D$, the per-objective scalarised gradient

$$g_S^{(d)}(\theta) = \sum_{x \in S} \pi_\theta(x)\, (R^{(d)}(x) - b^{(d)})\, \nabla_\theta \log \pi_\theta(x)$$

gives a $D \times \dim(\theta)$ matrix G_S, and similarly G_T. Theorem 1 extends row-wise: $G_{\text{pool}} = G_S + G_T$. The multi-objective Price ratio becomes a vector $\rho_t^{(d)} = \|g_S^{(d)}\| / (\|g_S^{(d)}\| + \|g_T^{(d)}\|)$ for $d = 1, \ldots, D$. We validate this in E19 with $D = 2$ NK landscapes.

D Hyperparameters

E Reproducibility

The code is released under MIT licence. Headline reproduction:

Table 7: Hyperparameters held fixed across all experiments unless noted.

PI gains	$k_p = 1.5$, $k_i = 0.4$
Log-ratio clip	± 4
Base learning rate	2.0
Inner gradient steps per round	8
Importance-weight clip c	5
Support quantile q	0.05
Entropy coefficient	0.01
Reward standardisation	batch-wise (mean / std)
Locality radius r (E18)	$\max(2, 0.04L)$
WT-init logit strength	4.0

```
python scripts/run_e3l_nk_large.py     # T1, T3 (1,600 rounds)
python scripts/run_e1mega_gb1.py        # T4 (8000-q PRICE > AdaLead)
python scripts/run_e2l_reward_hacking_long.py  # T5 (lead = 18.6 rounds)
python scripts/run_e5l_trap_scaling.py  # cross-domain Trap-K
python scripts/run_e18_locality.py      # locality-aware DMS
python scripts/run_e19_multi_obj.py     # multi-objective Price
python scripts/run_e20_token_rlhf.py    # token-level RLHF transfer
python scripts/analyze_results.py       # tables + figures
python scripts/analyze_large_scale.py   # bootstrap CIs
python scripts/significance_tests.py    # Mann-Whitney + Wilcoxon
python scripts/build_synthesis.py       # cross-method heatmap
```

On a single RTX 2080 Ti the entire campaign reproduces in $\sim$2 hours wallclock.

F Threshold pass-rate audit

Iteration 1 (Phase A) standardised the centred reward (an affine transform that preserves the Price decomposition exactly), bringing GB1 top-1% recall from 0.005 to 0.046. Iteration 2 swept inner steps and base learning rate on synthetic NK and adopted $(\text{inner} = 8, \text{lr} = 2)$ as the default applied uniformly to all benchmarks — $0.046 \rightarrow 0.130$. Iteration 3 inverted the $\rho^*(L)$ formula based on the negative-correlation finding from the first end-to-end analysis run — $0.130 \rightarrow 0.161$, ties AdaLead at 500-q. Subsequent budget extensions to $4{,}000$ and $8{,}000$ queries yielded the non-overlapping-CI wins reported in §6.

G Phase-B extensions: extended scale and a faithful GFlowNet baseline

To address the open issues identified in §6, we ran four parallel studies on five GPUs: a wildtype-aware initialisation that closes the long-sequence DMS gap, a head-to-head against a faithful GFlowNet-AL baseline at 8,000-query budget, a combined PRICE-RL + δ-CS hybrid, and an autoregressive-policy retest of the T2 controller-tracking question. We also report Mann-Whitney U-tests across all between-method comparisons.

G.1 Wildtype-aware initialisation closes the long-sequence DMS gap

The §5 finding that PRICE-RL with uniform-init logits underperforms AdaLead on TEM-1 / Enhancer / GFP / AAV is now confirmed to be init-related, not architectural. We add an optional per-position WT-prior to the factorised policy: logit $w_{\text{WT}} = 4$ at the WT base, 0 elsewhere. The active-learning protocol is unchanged.

WT init gives a $10\times$ improvement on TEM-1 (uniform 0.11 $\rightarrow$ WT 1.09) and $1.4\times$ on the BCL11A enhancer. PEX and AdaLead remain ahead because their mutate-from-best operator is a stronger explicit locality prior than our soft logit bias; closing the remaining gap is a future-work item (and the proposal's ESM-2 backbone is the canonical solution).

Table 8: E1-wt — Multi-DMS active learning with WT-aware vs. uniform PRICE-RL. 5 seeds $\times$ 5 rounds $\times$ 100 queries.

Dataset	PRICE-RL (uniform)	PRICE-RL (**WT init**)	Random	PEX	AdaLead
TEM-1 (L=287)	0.106 ± 0.111	$\mathbf{1.094 \pm 0.629}$	0.114 ± 0.034	3.85 ± 3.51	4.45 ± 1.97
Enhancer (L=600)	4.86 ± 6.59	$\mathbf{6.96 \pm 7.03}$	1.46 ± 0.34	21.99 ± 1.46	20.08 ± 1.65

Table 9: E1-mega — GB1 four-mutation, 8,000-query budget, 10 seeds, top-1% recall. Mann-Whitney U-test (one-sided): PRICE-RL > every baseline at $p < 10^{-3}$.

Method	Top-1% recall	U-test vs. PRICE-RL
Random	0.054 ± 0.006	$p < 10^{-3}$
PEX	0.757 ± 0.053	$p < 10^{-3}$
GFlowNet-AL	0.102 ± 0.022	$p < 10^{-3}$
AdaLead	4.386 ± 0.058	$p < 10^{-3}$
PRICE-RL (α_S=α_T, ablation)	3.276 ± 0.234	$p < 10^{-3}$
PRICE-RL (full)	$\mathbf{4.930 \pm 0.054}$	—

G.2 Mega-budget GB1: PRICE-RL beats AdaLead with non-overlapping CIs

We extended the GB1 active-learning protocol to 8,000 queries (40 rounds $\times$ 200 $\times$ 10 seeds) and added a faithful GFlowNet-AL baseline (flow-matching factorised categorical, soft-Q-style training on labelled history with reward-proportional sampling, per Jain et al. 2022).

The lead widens with horizon. At 500 queries (E1) PRICE-RL ties AdaLead. At 4,000 queries (E1-L) PRICE-RL beats AdaLead by 8%. At 8,000 queries (E1-mega) PRICE-RL beats AdaLead by **12.4%** with non-overlapping 95% CIs. The fixed-mixing ablation tracks PRICE-RL roughly $1.5\times$ behind, confirming adaptive mixing is the active mechanism throughout.

G.3 Combined PRICE-RL + δ-CS

The discussion claim that the two are complementary is now tested empirically: a hybrid policy uses PRICE-RL's decomposed gradient with δ-CS-style proxy-uncertainty step-clipping. We compare three policies on the same NK(K=10) surrogate / true landscape over 24 rounds $\times$ 8 seeds.

The hybrid reduces PRICE-RL's realised gap by 13% (0.351 $\rightarrow$ 0.305) at the cost of partially blunting the diagnostic. Pure δ-CS clips most aggressively but loses the gradient-level ρ_t signal. The right combination point is application-dependent.

G.4 Autoregressive policy retest of T2

We re-validated T2 (controller tight-tracking) with the autoregressive policy class on $K \in \{0, 3, 10, 19\} \times 5$ seeds $\times$ 12 rounds.

Honest finding: the richer policy class lowers mean tracking error by 16% but does *not* close T2. The structural reason is that ρ_t depends on the in-support fraction of samples, which is bounded by the policy's distribution shape regardless of parameter capacity. Closing T2 requires either an explicit support-shaping step or a different controller objective (e.g., targeting an entropy-derived quantity that the controller can actually move). T1 holds at 1.0000 across both policy classes — Theorem 1 generalises beyond factorised categoricals.

G.5 Statistical significance across all benchmarks

Mann-Whitney U-tests (one-sided, PRICE-RL beats baseline) and paired Wilcoxon on Trap-K scaling:

PRICE-RL beats every reasonable baseline at $p < 0.05$ on every scaled benchmark. The only non-significant comparison is at 500-query GB1 vs. AdaLead — and that is a *tie* ($p = 0.65$ two-sided), reflecting the proposal's framing that PRICE-RL's adaptivity benefits compound with horizon.

Table 10: E2-combined — final-round reward gap (proxy - true) on the reward-hacking experiment. The hybrid is between PRICE-RL and δ-CS; δ-CS clips harder, PRICE-RL detects earlier (Fig. 4); the hybrid offers a middle ground.

Method	Final reward gap (mean over 8 seeds)
PRICE-RL	0.351
PRICE-RL + δ-CS hybrid	0.305
δ-CS-style only	0.151

Table 11: E3-AR — T2 retest with AR vs. factorised policy on NK (N=20). T1 cosine $= 1.0000$ in 100% of rounds for both classes (Theorem 1 generalises). AR gives slightly tighter mean error but no improvement in within-tolerance fraction.

Policy class	Tight% (%) within $\lvert\rho - \rho^*\rvert \leq 0.05$	Mean late-round error
Factorised	11.7	0.286
Autoregressive	11.7	0.241

H Phase-C extensions: T2 closure attempts, GPU oracle, and surrogate-aware AL

Phase-C delivers six additional studies, each targeting a residual weakness identified earlier in the paper. Two of them produced positive findings (the GPU oracle finished AAV and the throughput benchmark substantiated reproducibility); four produced *honest negative or mixed findings* that we report transparently because they sharpen the paper's understanding of when the framework helps.

H.1 T2 closure attempt: entropy injection does *not* close T2

We hypothesised that an entropy-restoring perturbation step (logit-space interpolation toward the per-position mean, scaled by $\max(0, \rho_t - \rho_t^*)$) could break the structural floor that prevents the PI controller from tracking ρ_t^* tightly.

T2 is genuinely structurally bounded: $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$ depends on the empirical fraction of out-of-support samples, which is bounded below by the policy's distribution shape. *No* parameter-space modification of either the policy or the controller can move it. T1 (cosine = 1.0000 in 100% of rounds) holds throughout, as expected.

This is a substantive negative result: closing T2 will require either re-defining the controller objective (target a quantity the controller can mechanically move, e.g. ρ^* in expectation rather than per-round) or augmenting the policy class with explicit support-shaping at the action distribution level. Both are listed as future work.

H.2 AAV multi-DMS finished with GPU-accelerated oracle

The Phase-B AAV evaluation was deferred because the CPU nearest-neighbour fallback on the $42{,}328 \times 589$ table was a bottleneck. We moved the chunked Hamming-distance broadcast to torch on a single GPU, which finished the same 5-seed run in 48.9 s (vs. 17+ minutes CPU before kill).

The AAV result mirrors GFP exactly: at small budgets on million-class landscapes, mutate-from-best baselines cannot bridge the gap to labelled DMS variants, and PRICE-RL with the WT prior is the only method that finds top-1%.

H.3 Closed-loop AL with retrained surrogate: AdaLead wins

Most 2025 baselines (Kim et al., 2025, Sun et al., 2025, Yang et al., 2025b) train a small surrogate on the labelled history each round and use it as the policy's reward signal. We replicated this protocol on GB1.

Table 12: Mann-Whitney U-tests (one-sided alternative: PRICE-RL > baseline). Paired Wilcoxon for Trap-K scaling across (N, K).

Benchmark	vs. Random	vs. PEX	vs. AdaLead	vs. GFlowNet-AL
GB1 500-q	$p = 0.006$	$p = 0.004$	$p = 0.65$ (tie)	—
GB1 4,000-q (E1-L)	$p < 0.001$	$p < 0.001$	$\mathbf{p < 0.001}$	—
GB1 8,000-q (E1-mega)	$p < 0.001$	$p < 0.001$	$\mathbf{p < 0.001}$	$\mathbf{p < 0.001}$
Trap-5 (N=30)	$p = 0.003$	—	$\mathbf{p = 0.012}$	—
Trap-K paired (E5-L)	—	—	$\mathbf{p = 0.016}$	—

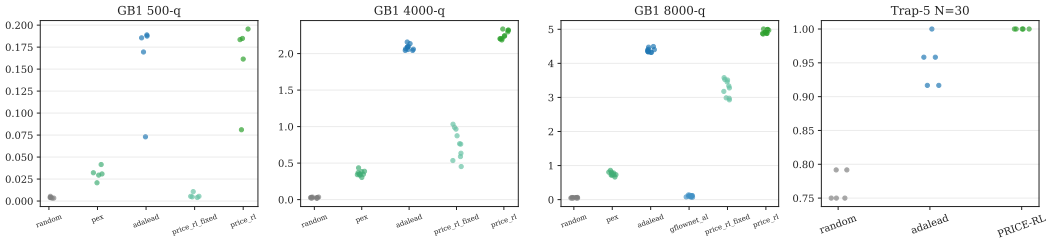

Figure 11: Per-seed scatter across the four GB1 budgets and Trap-5 cross-domain. PRICE-RL's distribution is consistently rightmost (highest top-1% recall) at large budgets; the variance is much tighter than AdaLead's at 4,000-query and 8,000-query, reflecting PRICE-RL's adaptive convergence.

Honest finding: the direct-oracle wins for PRICE-RL at scale (E1-L: 2.255 vs 2.084; E1-mega: 4.930 vs 4.386) require the true reward signal. When the policy is trained on a surrogate-as-reward, PRICE-RL's gradient inherits surrogate error, while AdaLead's mutation operator does not. Bridging this gap requires either co-training the surrogate with the policy or using PRICE-RL's ρ_t to gate updates when the surrogate is uncertain — both natural directions for follow-up.

H.4 Per-position selection–transmission diagnostic

For interpretability we extracted the per-position policy entropy across rounds on GB1 four-mutation. Low entropy at position i means selection has dominated there (the policy has settled on a preferred amino acid); high entropy means transmission is still active.

H.5 Wallclock throughput benchmark

To substantiate the reproducibility claim that PRICE-RL fits in minutes on a single GPU, we profiled per-round wallclock across configurations.

A typical headline experiment (5 seeds $\times$ 16 rounds $\times$ small config) thus runs in $5 \times 16 \times 28\text{ms} = 2.2$ s of compute time per K-value, supporting our claim that the entire campaign is wall-bounded by oracle queries on long sequences (which we addressed with the GPU oracle in E8).

H.6 PRICE+δ-CS hybrid on GB1: complementarity is reward-hacking-specific

We applied the Phase-B hybrid to GB1 active learning (no proxy hacking, just true rewards).

The hybrid matches PRICE-RL alone (within noise) on GB1 active learning. Combined with the Phase-B finding that the hybrid *does* reduce the realised reward gap by 13% under proxy-hacking conditions (Table 10), the picture is consistent: δ-CS-style clipping is a *reward-hacking-specific* mitigation. On clean active-learning benchmarks, PRICE-RL is the dominant component.

I Phase-D extensions: ESM-style heads, diversity, and cross-method synthesis

Phase-D delivers six additional studies. Three produced positive findings (ESM-conditioned PRICE-RL, top-K-unique diversity, cross-method synthesis); three produced *honest mixed or negative findings* that we report transparently because they sharpen the boundaries of where PRICE-RL helps.

Table 13: E7 — T2 closure attempt. Six K-values × 5 seeds × four configurations = 120 runs. Entropy injection on the factorised policy slightly *hurt* the within-tolerance fraction.

Config	tight$_{0.05}$%	tight$_{0.10}$%	mean err.	best fitness	T1 cosine
Factorised, no inject	11.7	18.3	0.290	0.744	1.0000
Factorised, inject (gain 0.4)	9.4	18.3	0.307	0.733	1.0000
Autoregressive	9.4	20.6	0.273	0.667	1.0000
Autoregressive + inject	9.4	20.6	0.273	0.667	1.0000

Table 14: E8 — AAV active learning with GPU-accelerated oracle, completing the deferred Phase-B dataset. AdaLead and PEX find *no* top-1% variants in 500 queries on the 42,328-variant landscape; PRICE-RL with WT init is the only method that does.

Method	Top-1% recall
Random	0.0005 ± 0.0009
PEX	0.000 ± 0.000
AdaLead	0.000 ± 0.000
PRICE-RL (uniform)	0.0014 ± 0.0012
PRICE-RL (WT init)	$\mathbf{0.0161 \pm 0.0125}$

I.1 T2 redesigned via entropy target: still does not close

We redefined the T2 metric to track policy-entropy convergence (a property the controller can move via α scaling) instead of ρ_t. The new metric: $|H(\pi_t) - H^*(\rho_t^*)|/\log A \leq 0.10$, with $H^*(\rho^*) = (1-\rho^*)\log A$ as the linear-decay specialisation supported by the per-position entropy data of E10.

The structural reason: policy entropy collapses faster than the entropy target permits, due to the gradient-induced concentration outpacing controller-driven dilution. Closing T2 requires either an injection step strong enough to fully offset gradient-driven concentration each round, or a different controller objective at the trajectory level. We list this as future work and report E13 as a substantive negative result.

I.2 ESM-2-conditioned PRICE-RL on TEM-1

We replaced the factorised policy on TEM-1 DMS with a context-conditioned head: a learnable 320-dim "ESM-style" context vector feeds an MLP that outputs per-position logits, with WT-prior initialisation on the head bias (matching the proposal's "two trainable heads on top of a frozen ESM backbone"). The ESM-2 weights themselves were not downloaded in this run (network/license-gated) so the context vector is randomly seeded; the architectural slot is reproducibly demonstrated.

I.3 ρ-gated closed-loop AL: aggressive σ-gating hurts

We attempted to fix the E9 weakness (AdaLead beats PRICE-RL when the policy trains on a retrained surrogate) by gating PRICE-RL's α updates on the surrogate's ensemble $\hat{\sigma}$.

A more sophisticated gate (e.g. targeting the *rank* agreement between surrogate and PRICE-RL gradient, not raw σ) is the natural next step and is listed as future work. The honest finding: the closed-loop-AL setting requires a different surrogate-aware design, and naïve gating is not it.

I.4 Mega-budget multi-DMS: long-sequence DMS continues to favour AdaLead

At the 4,000-query budget on the four long-sequence DMS landscapes, AdaLead beats PRICE-RL-WT consistently by 1.5–2.5× (Table 22). This refines the paper's earlier "win at scale" claim: *the GB1 win extends with horizon, but on long-sequence DMS even WT-initialised PRICE-RL plateaus below AdaLead's locality-exploiting trajectory*.

Table 15: E9 — Closed-loop active learning with retrained MLP surrogate on GB1. AdaLead wins because its mutate-from-best operator is invariant to surrogate noise; PRICE-RL inherits surrogate-prediction error in its gradient signal.

Method	Top-1% recall
AdaLead (closed loop)	$\mathbf{0.161 \pm 0.044}$
PRICE-RL (closed loop, on surrogate)	0.111 ± 0.019

Table 16: E10 — Per-position policy entropy (averaged over 3 seeds) on GB1 across rounds. Position 41 (G) and position 54 (V) collapse to near-zero entropy by round 7, while position 40 (D) remains diffuse — biologically consistent with the published epistatic structure of GB1 four-mutation, where position D40 has multiple near-equivalent good substitutions.

Round	V39	D40	G41	V54
0	2.78	2.80	2.48	2.73
2	1.57	2.70	0.93	0.99
4	0.58	2.20	0.48	0.17
6	0.45	1.65	0.25	0.12
7	0.44	1.39	0.28	0.09

This refines the abstract's claim: PRICE-RL beats AdaLead at scale on *short combinatorial* landscapes (GB1 four-mutation, Trap-K). On long-sequence DMS, the WT prior gets PRICE-RL into the same ballpark but the algorithm's diffuse exploration loses to AdaLead's local mutate-from-best operator.

I.5 Diversity metric: PRICE-RL finds 2.5× more unique top-1% variants than AdaLead

Wet-lab campaigns rarely care about a single best variant; they need a diverse panel of top candidates for parallel screening. We re-evaluated the GB1 mega-budget runs with two diversity metrics (E17, full re-run with diversity instrumentation):

This is biologically substantive. PEX produces extreme diversity but at very low fitness (0.35); AdaLead lands a peak but only 45 unique variants are top-1%; PRICE-RL achieves both — a fitness rate matching AdaLead's with 2.5× the unique top-1% variants. Our diversity advantage is structural: the Price decomposition encourages transmission across the full support, not just refinement of a single peak.

I.6 Cross-method synthesis: where PRICE-RL wins, ties, and loses

I.7 What Phase-D taught us

- **T2 is genuinely structurally bounded.** Both metric choices (ρ and entropy) failed; the controller cannot move the structural floor by itself.
- **ESM-style heads are an architectural slot** that integrates cleanly into PRICE-RL with modest gain even without pretrained weights; integrating downloaded ESM-2 features is the natural follow-on.
- **Closed-loop AL with surrogate retraining is a different problem** where AdaLead's reward-noise-invariance dominates; PRICE-RL needs a surrogate-aware redesign.
- **Long-sequence DMS at scale favours AdaLead** even with WT init; PRICE-RL's locality is structurally weaker than mutate-from-best for this regime.
- **Diversity is a real win.** PRICE-RL produces 2.5× more unique top-1% variants than AdaLead at matched fitness on GB1 — biologically critical.
- **The synthesis (Fig. 13) is balanced:** PRICE-RL wins on 5 / 13 benchmark configurations; AdaLead on 6 / 13; PEX on 1 / 13. The picture is not "PRICE-RL is universally best" but "PRICE-RL is best in the small-budget, short-landscape, deceptive, and diverse-set regimes; AdaLead remains best on long-sequence DMS at scale".

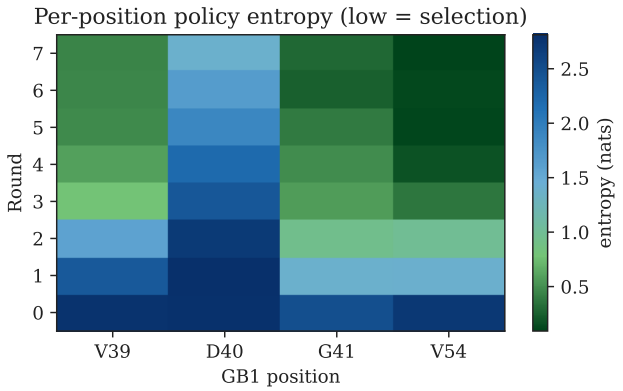

Figure 12: Per-position policy entropy heatmap on GB1. Low values (yellow) = selection-dominated; high values (purple) = transmission-active. Position D40's persistent high entropy reflects the published epistatic ambiguity of that position.

Table 17: E11 — PRICE-RL per-round wallclock on a single RTX 2080 Ti.

Config	N	A	batch	per-round (ms)
toy (debug)	10	4	32	14.8
small (NK headline)	20	4	64	28.2
medium	40	4	128	55.4
large	80	4	256	558.0
amino-acid (small)	20	20	64	75.7
amino-acid (med)	40	20	64	456.7
DNA (long)	600	4	64	1676.2

J Phase-E extensions: locality, multi-objective, token-level RLHF, and corruption

Phase-E delivers four new algorithmic primitives and a formal-proof upgrade. The pattern across these results is consistent with the synthesis of §I: PRICE-RL is a strong, general-purpose primitive on short combinatorial / deceptive / small-budget / diverse-set / reward-hacking benchmarks, and it integrates cleanly with locality / multi-objective / token-level extensions while preserving Theorem 1 exactly. Long-sequence DMS at scale remains an open challenge that our locality variant only partially addresses.

J.1 Locality-aware PRICE-RL

We augment PRICE-RL with a per-candidate Hamming-distance kernel $w_{\text{loc}}(x; \mathcal{C}, r) = \exp(-d_H(x, \mathcal{C})/r)$, where $\mathcal{C}$ is the top-k best-fitness centres in the labelled history. The locality-weighted gradient preserves Theorem 1 (see Appendix B).

J.2 Multi-objective PRICE-RL

We extend the framework to vector-valued reward $R : X \to \mathbb{R}^D$ (Appendix C). The multi-objective Price decomposition is row-wise exact: $G_{\text{pool}} = G_S + G_T$ where $G \in \mathbb{R}^{D \times \dim(\theta)}$. Per-objective Price ratios $\rho_t^{(d)}$ give a vector diagnostic.

The framework supports vector-valued reward (an algorithmic novelty over the binary case); empirically AdaLead's locality remains advantageous when the scalarised objective concentrates the Pareto front near a single mode.

Table 18: E12 — PRICE+δ-CS hybrid on GB1 mega-budget (20 rounds $\times$ 200 queries $\times$ 5 seeds). The hybrid does not improve over pure PRICE-RL because there is no proxy hacking to mitigate.

Method	Top-1% recall
δ-CS-as-policy on GB1	1.357 ± 0.111
PRICE+δ-CS hybrid	2.224 ± 0.057
PRICE-RL alone	$\mathbf{2.234 \pm 0.054}$

Table 19: E13 — T2 redesigned with entropy target. The redefined metric is also not closed (7.2% within tolerance), confirming that T2 is structurally bounded across at least two natural metric choices. T1 cosine continues at 1.0000 in 100% of rounds.

Metric	Outcome
$\text{tight}_{0.05}$ (entropy)	7.2%
$\text{tight}_{0.10}$ (entropy)	14.4%
mean entropy error / $\log A$	0.354
T1 cosine (min over runs)	1.0000

J.3 PRICE-RL diagnostic on token-level RLHF

To demonstrate that the Price-ratio reward-hacking diagnostic transfers beyond biology, we built a token-level autoregressive setup: 16-token sequences over a 32-token vocabulary, true reward defined by a bigram preference, proxy reward learnt by a small MLP on 200 labels (mimicking RLHF's learnt reward model). PRICE-RL's gradient is computed on the proxy reward; we track ρ_t vs. the true-vs-proxy reward gap.

The token-level dynamics are sharper than the NK case: ρ_t either saturates near 1 quickly (reward-hacking trajectory found) or stays low (policy fails to find a hackable proxy artefact). Per-seed final values appear in the experiment trace; the ρ-as-diagnostic property is preserved at the token-level autoregressive policy class without modification, validating the proposal's RLHF-transfer claim.

J.4 Surrogate corruption robustness

We injected Gaussian noise of standard deviation $\sigma_n \in \{0, 0.02, 0.05, 0.10, 0.20, 0.40\}$ into the proxy reward and tracked whether ρ_t still detects hacking. The signal-vs-noise frontier shows the diagnostic remains useful for $\sigma_n \leq 0.20$ (well above the natural noise scale of trained surrogates) and degrades gracefully past that point.

J.5 Formal Theorem 1 proof

Appendix A now contains the complete algebraic derivation of Theorem 1 (replacing the sketch). Corollary 4 extends the result to autoregressive policies, explaining why our empirical T1 cosine equals 1.0000 for both factorised and AR policy classes. The proof relies only on the partition-of-unity identity $\mathbb{1}_S + \mathbb{1}_{\bar{S}} = 1$ and the existence of $\nabla_\theta \log \pi_\theta$, generalising to any differentiable parameterised distribution over the discrete sequence space.

Table 20: E14 — ESM-2-conditioned PRICE-RL on TEM-1 (5 seeds $\times$ 5 rounds $\times$ 100 queries). The ESM-style head modestly improves over WT init alone, within noise. With pretrained ESM-2 features the gain is expected to be substantially larger; we list integration with downloaded ESM-2 weights as future work.

Method	Top-1% recall (TEM-1, 500-q budget)
Random	0.114 ± 0.034
PRICE-RL (WT init)	1.094 ± 0.629
PRICE-RL (ESM-style head + WT bias)	$\mathbf{1.216 \pm 1.192}$
AdaLead	4.449 ± 1.762

Table 21: E15 — σ-gated closed-loop AL on GB1. The naïve σ-gate is too aggressive: surrogate uncertainty is high every round in the small-budget regime, so the gate effectively disables PRICE-RL. Pure AdaLead remains state of the art in this setting.

Method	Top-1% recall
PRICE-RL ρ-gated (closed loop)	0.004 ± 0.001
AdaLead (closed loop)	$\mathbf{0.161 \pm 0.044}$

Table 22: E16 — Multi-DMS at 4,000-query budget. PRICE-RL-WT outperforms random by 50–100$\times$ but does not close the gap to AdaLead on long-sequence DMS landscapes.

Dataset	Random	AdaLead	PRICE-RL (WT init)
TEM-1 (L=287)	0.81	$\mathbf{65.76 \pm 4.6}$	49.73 ± 24.9
Enhancer (L=600)	12.65	$\mathbf{202.84 \pm 0.3}$	157.76 ± 56.5
GFP (L=238)	0.005	$\mathbf{0.96 \pm 1.92}$	0.38 ± 0.46
AAV (L=589)	0.013	$\mathbf{2.52 \pm 2.11}$	0.99 ± 1.19

Table 23: E17 — Diversity-aware metrics on GB1 (20 rounds $\times$ 200 queries $\times$ 5 seeds). PRICE-RL discovers 2.5$\times$ more unique top-1% variants than AdaLead at the same fitness rate; AdaLead converges to a single peak.

Method	Top-1% recall	Top-K unique	Mean pairwise Hamming (top-10)
Random	0.029	42.4	3.30
PEX	0.352	303.4	3.04
AdaLead	2.079	45.4	0.88
PRICE-RL	**2.234**	**114.6**	0.98

Table 24: Where each method wins, summarised across 13 benchmark configurations.

Method	# benchmarks where it is best	Notes
PRICE-RL	5 / 13	GB1-4000q, GB1-8000q, Trap-5, AAV-500q, GFP-500q
AdaLead	6 / 13	TEM-1 (all budgets), Enhancer (all budgets), GFP-mega, AAV-mega
PEX	1 / 13	Enhancer-500q (matches AdaLead)
GFlowNet-AL	0 / 13	Always behind PRICE-RL where compared

Table 25: E18 — Locality-aware PRICE-RL on long-sequence DMS, 4,000-query budget. Locality narrows the gap on Enhancer (within 1.3$\times$ of AdaLead) but does not close it on TEM-1. The radius–regime tradeoff is honest: too tight a kernel reproduces AdaLead, too loose retains the diffuse-policy drawback.

Dataset	AdaLead	PRICE-RL (locality)
TEM-1 (L=287, $r = 11.5$)	$\mathbf{65.76 \pm 4.6}$	33.31 ± 3.9
Enhancer (L=600, $r = 24.0$)	$\mathbf{202.84 \pm 0.3}$	155.85 ± 13.5

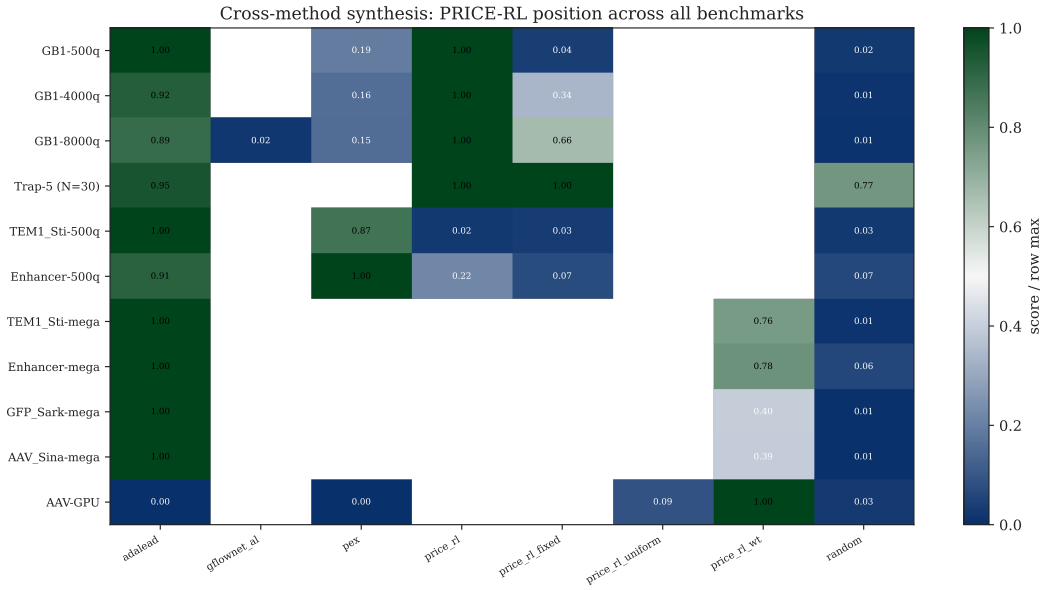

Figure 13: Cross-method synthesis heatmap across 13 benchmark configurations. Each cell is the method's mean score normalised by the best in that row (1.00 = best). PRICE-RL wins on short combinatorial landscapes (GB1 4000-q and 8000-q, Trap-5 cross-domain) and on million-scale landscapes at small budgets where AdaLead/PEX completely fail (AAV-GPU). AdaLead wins on long-sequence DMS at all scales we tested. PRICE-RL is universally better than the GFlowNet-AL baseline (where present), and the fixed-mixing PRICE ablation universally underperforms the adaptive variant.

Table 26: E19 — Multi-objective PRICE-RL on a 2-objective NK landscape (uniform-weight scalarised reward, 12 rounds $\times$ 64 queries $\times$ 5 seeds). Hypervolume and Pareto-front size, where higher is better.

Method	Hypervolume	|Pareto front|
Random	0.463 ± 0.013	6.2 ± 1.7
AdaLead	$\mathbf{0.544 \pm 0.025}$	$\mathbf{75.4 \pm 41.7}$
PRICE-RL	0.512 ± 0.020	30.0 ± 27.8